

Discussion 9

Chapter 1

- 1) What mathematical concepts are used to describe the structure of any crystal?

We use a set of vectors to describe the repeating nature of a Bravais lattice

$$\vec{R} = m\vec{a}_1 + n\vec{a}_2 + o\vec{a}_3$$

The basis of the Bravais lattice does not necessarily contain only one atom. They may be two atoms or three atoms instead.

- 2) What are the typical crystal structures of metals and why are they common?

The typical crystal structure are fcc and hcp. The close packed structure maximizes the overlap between the atomic wave functions and therefore maximizes the interaction energy. Also because metals have no directional bonding

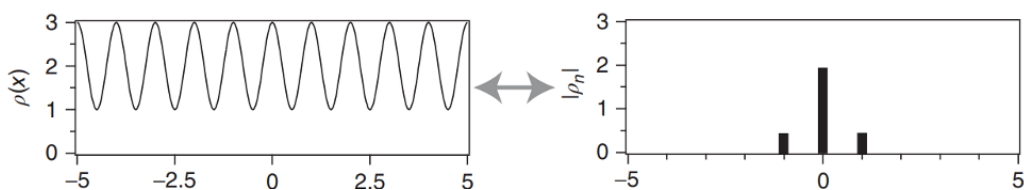
- 4) How can the reciprocal lattice conveniently be used to describe lattice-periodic functions?

While the lattice describes the position of our atoms, the reciprocal lattice describes the periodicity. This is a result of the Fourier transform. Each lattice point in the reciprocal lattice describes a periodicity in the lattice, which we can describe by Miller indices. This is what makes up the Miller planes.

When describing a real lattice then we need to store a whole function $\rho(x)$ on the computer. That's a lot of number. When converting it to a reciprocal lattice you only store the period of that function which is a lot more efficient.

Let's take the example below.

For the real lattice you would have to store all the numbers for at least one period. The reciprocal lattice is a representation of the periodicity so you would only have to store 2 numbers in this case (the offset and the period of the cosine wave). The efficiency of the reciprocal lattice representation storing the Fourier coefficient is a lot better than storing all the numbers of the real space lattice



7) How can the reciprocal lattice of a crystal be used to predict the pattern of diffracted X-rays?

From diffraction theory, the Laue condition says that we only see intensity when the scattering vector is equal to the vectors in the reciprocal lattice. Therefore when we do diffraction experiments, what we see in the images of a diffraction pattern are the points making up the reciprocal lattice. Hopefully, you imagine how taking the inverse Fourier transform should yield the real space lattice

Chapter 2

1) Why is a typical interatomic potential, such as the one in Figure 2.1, asymmetric?

The repulsion is due to Pauli exclusion principle. It acts like a hard shell because the wave function of overlapping atoms must be orthogonal so they must enter higher and higher energy states as they close the distance. The energy should approach infinity

The attraction is due to Coulomb attraction which asymptotes at zero energy

The effects of repulsion and attraction come from different sources which is why it is antisymmetric. Also the interatomic potential reaches different values in the limit of $r \rightarrow 0$ and $r \rightarrow \infty$

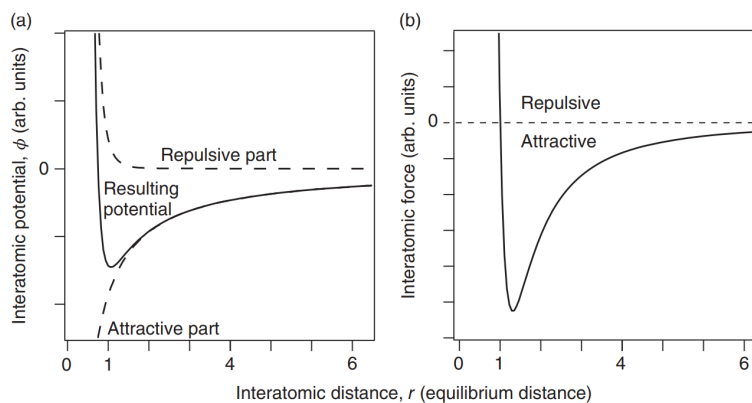
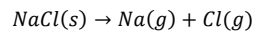


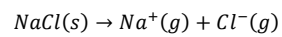
Figure 2.1 (a) Typical interatomic potential $\phi(r)$ for bonding in solids according to (2.1) with $n = 6$ and $m = 1$. (b) Resulting force, that is, $-\text{grad}\phi(r)$.

5) Explain the difference between cohesive energy and lattice energy.

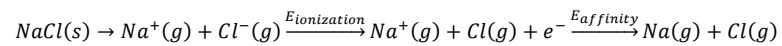
The cohesive energy is the total energy difference between any solid and the atoms that it is made of



The lattice energy is the total energy to separate the solid into free gaseous ions



Therefore the cohesive energy is the lattice energy plus the electron affinity to turn $Na^+ \rightarrow Na$ and the ionization energy to turn $Cl^- \rightarrow Cl$



Discussion 10

6) Which elements are likely to form metals?

Transition metals are likely to form metals

8) What is the difference between a simple metal and a transition metal (definition and typical physical properties)?

A simple metal is one where conduction electrons come from s and p orbitals

Simple metals are typically **lustrous**, **malleable** (can be hammered into sheets), and **ductile** (can be drawn into wires). They are also excellent **conductors of heat and electricity**, have a **high melting point**, and are generally **solids at room temperature** (with exceptions like mercury).

A transition metal is one where conduction electrons come from d orbitals

Transition metals are typically hard, dense, and shiny with high melting and boiling points, making them excellent conductors of heat and electricity. They are also malleable and ductile, meaning they can be hammered into shape or drawn into wires.

Physical properties

Property 	Transition Metals	Simple Metals
Hardness and strength	Hard and strong	Soft
Melting point	High	Low
Density	High	Low
Conductivity	Good conductors of heat and electricity	Good conductors of heat and electricity

1) What shape does the phonon dispersion, that is, the vibrational frequency as a function of wave vector $\omega(k)$, take for an infinite chain of atoms with one atom per unit cell?

An infinite chain of monoatomic atoms have the following phonon dispersion

$$\omega(k) = \sqrt{\frac{2\gamma(1 - \cos(ka))}{M}} = 2\sqrt{\frac{\gamma}{M}} \left| \sin\left(\frac{ka}{2}\right) \right|$$

Graphically it looks as below

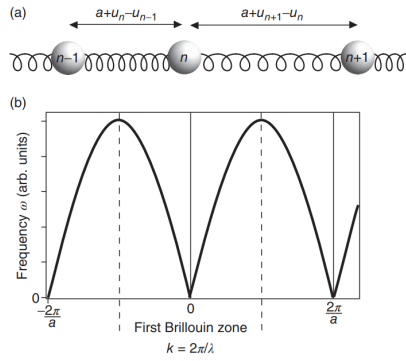


Figure 4.1 (a) One-dimensional chain with one atom per unit cell. (b) Allowed vibrational frequencies ω as a function of the wave vector k .

Discussion 11

- 2) What shape does take for two atoms per unit cell? Why does one speak of optical and acoustic branches?

The optical branch allows optical excitation of the $k = 0$ mode

The acoustic branch has a dispersion relation at $k = 0$ that is determined by the speed of sound

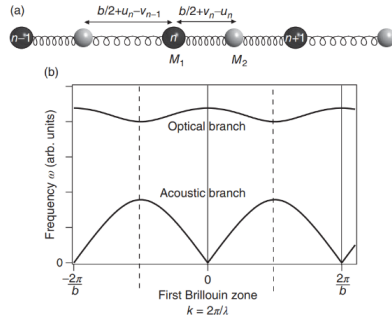


Figure 4.4 (a) One-dimensional chain with two atoms per unit cell. (b) Allowed vibrational frequencies ω as a function of the wave vector k .

- 3) The phonon dispersion for a one-dimensional chain with one and two atoms per unit cell is given by Eqs. (4.10) and (4.17), respectively. What about the amplitude of the vibrations?

The amplitude of the vibrations for a one atom per unit cell chain is derived from the equipartition theorem

$$\langle E \rangle = \left\langle \frac{1}{2} m v^2 + \frac{1}{2} \gamma x^2 \right\rangle = kT$$

When the kinetic energy is zero then the displacement is at its maximum. This is exactly what the amplitude is

$$x_m = \sqrt{\frac{2kT}{\gamma}}$$

For a two atom per unit cell it is

$$\begin{aligned} -M_1 \omega^2 u &= -2\gamma u + \gamma v (e^{-ikb} + 1) \\ -M_2 \omega^2 v &= -2\gamma v + \gamma u (e^{ikb} + 1) \end{aligned}$$

5) Explain the meaning and use of periodic boundary conditions to describe the properties of finite solids.

We want to ignore any surface interactions and any problems that may come with it under the assumption that it will not affect the system that much

1) Describe the basic assumptions of the Drude model for metals. Explain the relaxation time and the mean free path of the electrons.

The Drude model is based on the model of classical gas and charged particles

1. Independent electron approximation where electrons in a solid do not interact with each other
2. Free electron approximation where electrons do not interact with ion cores in between collisions
 - The ion cores are immobile
 - The collisions instantaneously change the velocity of the electron
3. The electrons reach thermal equilibrium with the lattice through collision with the ions

The relaxation time is the time between collisions

The mean free path is average length where the electrons do not collide with anything

2) How fast do the electrons move in the Drude model, and how does their speed depend on temperature?

The kinetic energy is connected with temperature through the equipartition theorem

$$\frac{1}{2}mv^2 = \frac{3}{2}kT$$

At room temperature they move with a speed $v_t \approx 10^5 \frac{m}{s}$

3) Describe the electrical conduction in the Drude model. What is the origin of electrical resistance?

Electrical conduction is based on the drift velocity in the presence of an electric field

The electrical resistance arises because of electrons colliding with ion cores as given by the finite mean free path

7) What is the Hall effect, and what can be measured with its aid?

The Hall effect is a phenomenon where an electric field is built up upon exposure to a magnetic field

The conduction electron density can be measured by measuring the Hall coefficient

$$\mathcal{E}_H = R_H j_x B_z \Rightarrow R_H = -\frac{1}{ne}$$

Discussion 12

9) Why do metals not transmit light? Is this true for all light frequencies?

The reflectivity depends on the plasma frequency. Metals will reflect low frequency light when the frequency is lower than the plasma frequency. But when the frequency of light is larger than the plasma frequency then the metal will be transparent

11) Which properties of metals are not described adequately by the Drude model?

A list of shortcomings

- The mean free path should increase with decreased temperature
- The wrong Wiedemann-Franz law
- The electronic contribution to heat capacity is too large
- The resistivity of alloys can be much larger than the components
- The Hall coefficient can be positive
- Wrong magnitude of conductivity at low temperature
- Deviancy in Hall coefficient of Bi

3) How do the free electrons in the quantum model contribute to the heat capacity and how does this differ from the Drude model?

All electrons contribute to the heat capacity in the Drude model

Only electrons near the Fermi level contribute in the quantum model

4) How does the contribution of the electrons to the heat capacity depend on temperature?

The electronic contribution to the heat capacity is given by

$$C_V = \frac{\pi^2}{3} k_B^2 T g(E_F) \propto T$$

6) What is the form of the general solution of the Schrödinger equation for a lattice-periodic potential?

It is given by Bloch's theorem. It is a plane wave modified by a lattice-periodic function

$$\psi_{\vec{k}}(\vec{r}) = e^{i\vec{k} \cdot \vec{r}} u_{\vec{k}}(\vec{r})$$

$$u_{\vec{k}}(\vec{r}) = u_{\vec{k}}(\vec{r} + \vec{R}) = \sum_{\vec{G}} c_{\vec{k}-\vec{G}} e^{-i\vec{G} \cdot \vec{r}}$$

14) What is the physical interpretation (or several interpretations) of the vector $\mathbf{k}$ for an electronic state in a solid?

The k-vector is the quantum number of the electronic states that the electrons can exist in

$$\vec{k} = (k_x, k_y, k_z) = \frac{2\pi}{L} (n_x, n_y, n_z)$$

The vector participates in the wavefunction to describe the electrons as plane waves

$$\psi_{\vec{k}}(\vec{r}) = e^{i\vec{k} \cdot \vec{r}} u_{\vec{k}}(\vec{r})$$

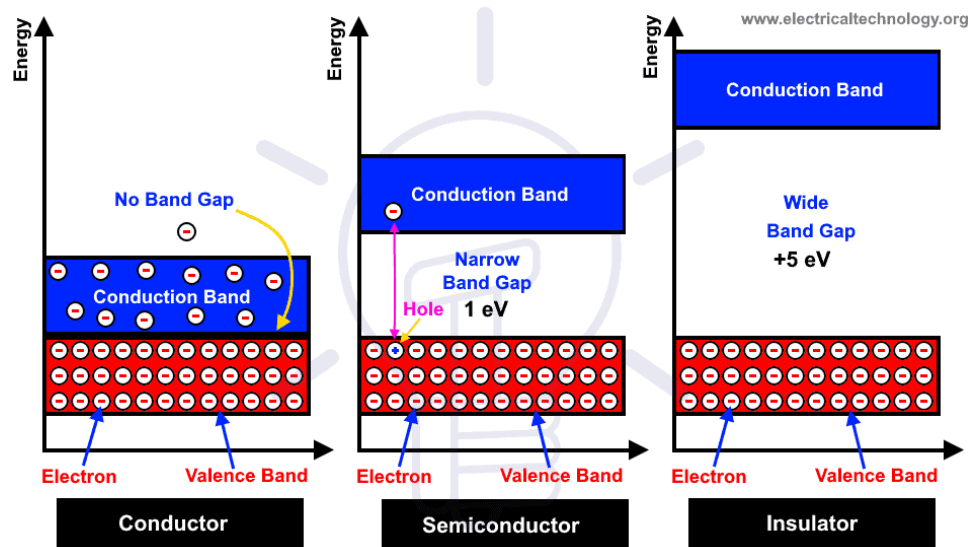
We can therefore also consider the vector as describing the direction and frequency of the electrons

Discussion 13

1) What is the difference between a metal, a semiconductor, and an insulator?

They are determined by their band gap.

- Metals do not have a band gap
- Semiconductors have a small band gap. Excitation from the valence band to conduction band is easily possible through thermal or optical excitations
- Insulators have a large band gap. Excitations are very difficult to achieve



2) How does the conductivity of a semiconductor (typically) change as a function of temperature and why?

The conductivity depends if the semiconductor is doped or not. In the intrinsic case it behaves like this

$$\sigma \propto n_i = p_i = \sqrt{np} = 2 \left(\frac{kT}{2\pi\hbar^2} \right)^{\frac{3}{2}} (m_e^* m_h^*)^{\frac{3}{4}} e^{-\frac{E_g}{2kT}} \propto T^{\frac{3}{2}} e^{-\frac{E_g}{2kT}}$$

$$\Downarrow$$

$$\ln(\sigma) \propto \ln \left(T^{\frac{3}{2}} e^{-\frac{E_g}{2kT}} \right) = \frac{3}{2} \ln T - \frac{E_g}{2kT}$$

The conductivity increases with temperature as the electrons are excited from the VB to CB. If the semiconductor is doped then the conductivity will increase in two steps. The first one involves excitation of the donor level. The second one involves excitation of the VB

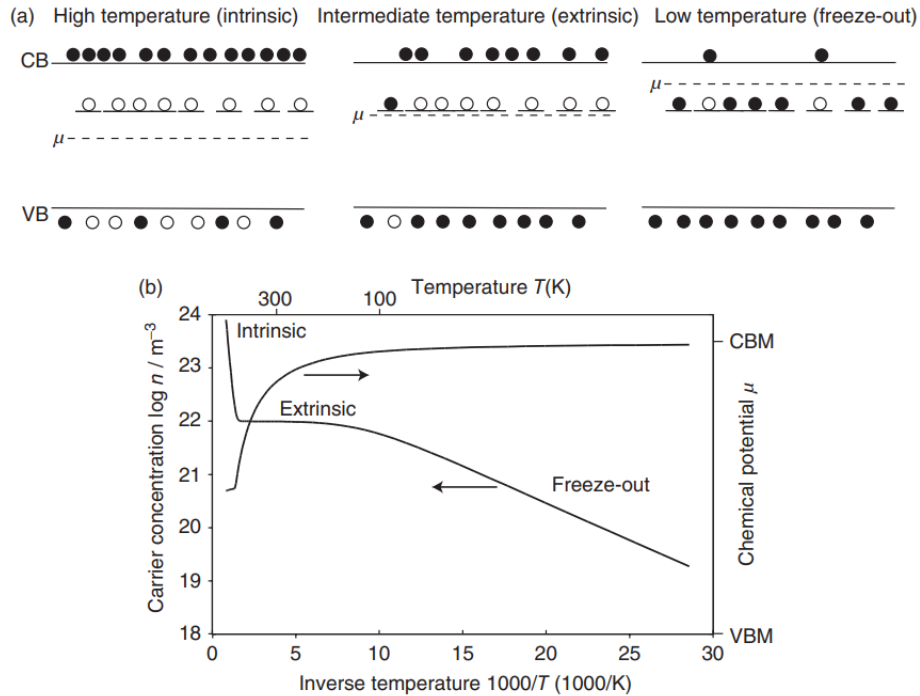


Figure 7.7 Electron density and position of the chemical potential for an n-doped semiconductor. (a) *Qualitative picture*: At very low temperatures, almost none of the donors are ionized (freeze-out region). At intermediate temperatures, almost all of the donors are ionized but very few intrinsic carriers are excited (extrinsic region). At high temperatures, all the donors are ionized and some intrinsic carriers are excited (intrinsic region). (b) *Quantitative picture*: The position of the chemical potential and the logarithmic carrier density are shown as a function of the inverse temperature.

4) Explain the difference between “electrons” and “holes.”

Electrons are the particles as we know them.

Holes are quasiparticles to describe the absence of electrons. They behave like particles but are technically not. The concept is similar to that of phonons

6) What is the meaning of the effective mass, and how can it be related to the electronic band dispersion?

The effective mass is given by

$$m^* = \hbar^2 \left(\frac{d^2 E(k)}{dk^2} \right)^{-1}$$

It describes the curvature of a band. The reason we use effective mass is such that we can use quasiclassical theory. In the book they derived this equation from the group velocity

$$a = -\frac{1}{\hbar^2} \frac{d^2 E(k)}{dk^2} e\mathcal{E}$$

This is very close to Newton's second law

$$F = -e\mathcal{E} = ma$$

But in order to fulfill this equation we have to define the effective mass as $\hbar^2 \left(\frac{d^2 E(k)}{dk^2} \right)^{-1}$. It is therefore a way to simplify a lot of calculations in the future by using physics we already know rather than introducing new and more complicated physics

10) Why does a very small number of donor or acceptor atoms in a semiconductor have a big impact on the concentration of free carriers?

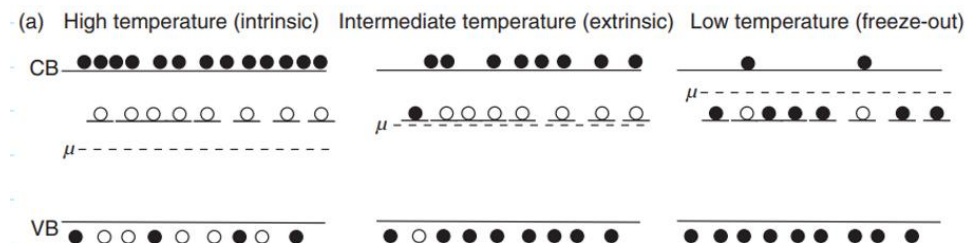
The intrinsic carrier concentration is very small. Typical doping concentration is in the order of 10^{20} m^{-3}

Table 7.2 Intrinsic carrier densities for Si and GaAs.

Material	Gap size (eV)	n_i at 150 K (m^{-3})	n_i at 300 K (m^{-3})
Si	1.11	4.1×10^6	1.5×10^{16}
GaAs	1.43	1.8×10^0	5×10^{13}

12) Where does the chemical potential in an n-doped semiconductor lie at low and high temperatures?

It is as given in the picture



13) Explain why a pn junction works as a check valve for the current.

A check valve is a one-way valve, so current only goes one way.

A pn junction is created when a p-material is in contact with an n-material. Electrons from the n-side combine with holes from the p-side which eventually creates a depletion zone associated with a band bending. The depletion zone will contain an electric field that resists any more flow of electrons from the n-side to the p-side. This electric field is what works as a one-way valve. To open the valve we have to apply an opposite electric field counteract the one that occurs naturally in the material

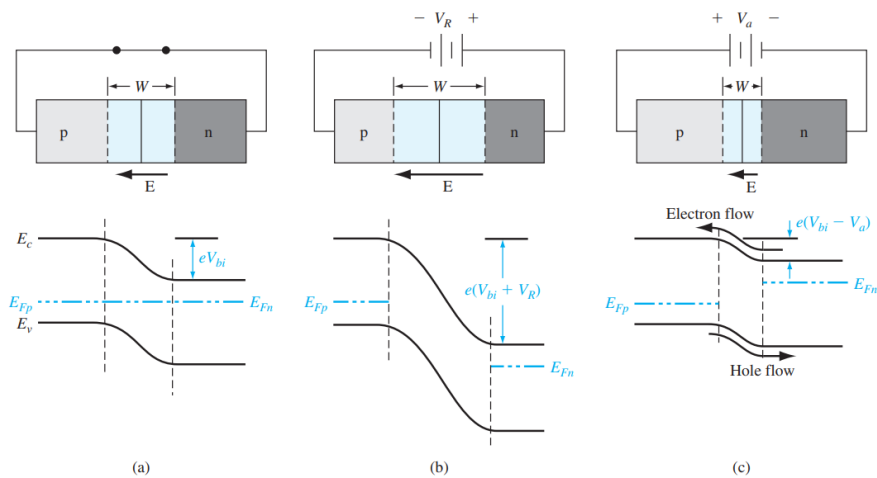


Figure 8.1 | A pn junction and its associated energy-band diagram for (a) zero bias, (b) reverse bias, and (c) forward bias.

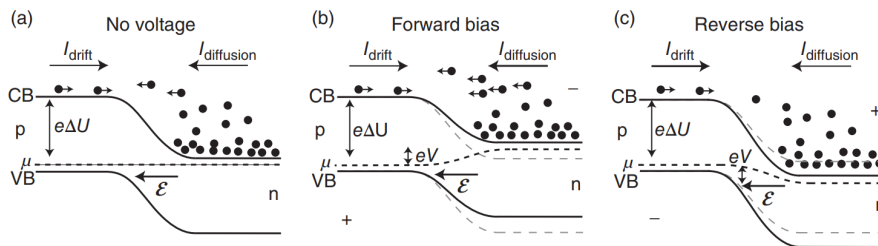


Figure 7.11 The pn junction as a diode (only considering the electrons, not the holes). (a) Without any bias voltage. (b) Forward bias. (c) Reverse bias. The gray dashed lines show the position of the n-side VB and CB without an applied voltage.

Discussion 14

- 1) What is the difference between diamagnetism and paramagnetism? What are the basic physical causes for these phenomena? How do diamagnetic/paramagnetic materials react when placed into an inhomogeneous magnetic field?

The main difference is the magnetic susceptibility

- Diamagnetism has $\chi_m < 0$ which means that a diamagnetic material will be repelled by a magnetic field
- Paramagnetism has $\chi_m > 0$ which means that a paramagnetic material will be attracted to a magnetic field

The basic physical cause is spin-pairing

- If all electrons are paired with opposite spins then the magnetic moments cancel and you have diamagnetism
- If some electrons are unpaired then they have the option to align with an external magnetic field and you have paramagnetism

- 4) How do the free electrons in metals contribute to the magnetic properties?

Free electrons are described by Pauli paramagnetism

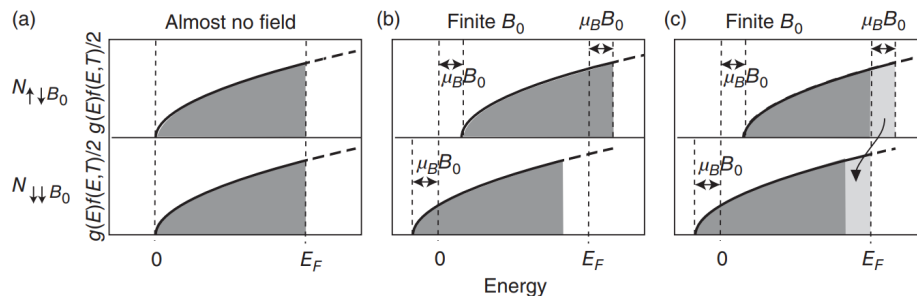


Figure 8.3 (a) Density of occupied states for free electrons at $T = 0$ K, split up into electrons having their magnetic moment antiparallel ($N_{\uparrow B_0}$) or parallel ($N_{\downarrow B_0}$) to an external field, but the field is almost zero. (b) When B_0 is no longer small, the energy of the electrons is increased or reduced by $\mu_B B_0$, depending on the orientation of their

magnetic moments. (c) The electrons with a magnetic moment antiparallel to the field can reach a lower energy state by flipping their spin. In this way, a stable situation with a constant Fermi energy is reached. Note that the energy shift induced by B_0 is not drawn to scale.

- 6) In a Curie-type paramagnet, all ions with magnetic moments contribute to the total susceptibility, but in a metal, only very few of the free electrons contribute to Pauli paramagnetism. Why?

Only electrons around the Fermi level contribute to the magnetism. Look at the picture above as to why

8) What interaction is responsible for the magnetic ordering in ferromagnets and antiferromagnets and how strong is it?

Magnetic ordering is described from the exchange energy

$$E_{\uparrow\uparrow} - E_{\uparrow\downarrow} = -2X$$

We can describe the exchange energy using the Weiss field which is on the order of thousands of teslas

$$\vec{B}_W = \frac{n_{nn}XV}{g_e^2\mu_B^2N}\vec{M}$$

9) Explain the origin of the Weiss field and its significance for magnetic ordering.

The Weiss field arises from the mean field approximations where we look only in a local region of nearest neighbours and approximate that they have the same global average spin. This results in the Hamiltonian being written as a sum of two magnetic fields where the Weiss field is one of them. Note that the Weiss field is not an actual magnetic field but a way to describe the local interaction

1) What are the physical mechanisms that can lead to a dielectric polarization of solids?

There are three

1. Atomic polarization where the electron cloud can be shifted by an electric field
2. Ionic polarization where the ionic lattice can be distorted in the presence of an electric field
3. Oriental polarization where asymmetrically charged atoms can be oriented by an electric field

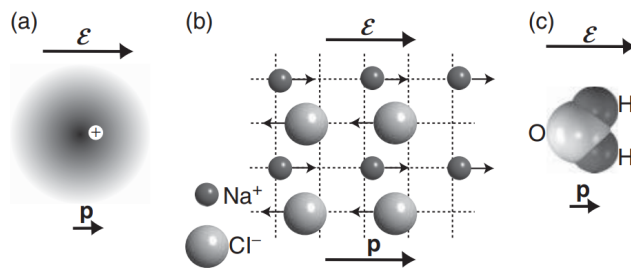


Figure 9.2 Mechanisms leading to microscopic electric polarization. (a) The electric field polarizes all the atoms in the solid. (b) In ionic solids, like NaCl, the lattice can be polarized, giving rise to local electric dipoles. The dashed grid gives the position of the

ions without an applied field. (c) If there are permanent dipoles in the solid and these are free to rotate, they orient themselves parallel to the field. A molecule with a permanent dipole is, for example, water.

3) Qualitatively describe the frequency dependence of the dielectric function for a material with both electronic and ionic polarization.

We should have two resonant frequencies now instead of one as shown in the book. One resonant frequency for the ionic polarization at low frequency and one resonant frequency for electronic polarization at high frequency. After each resonant frequency the dielectric function should be smaller because we “lose” more and more polarizability

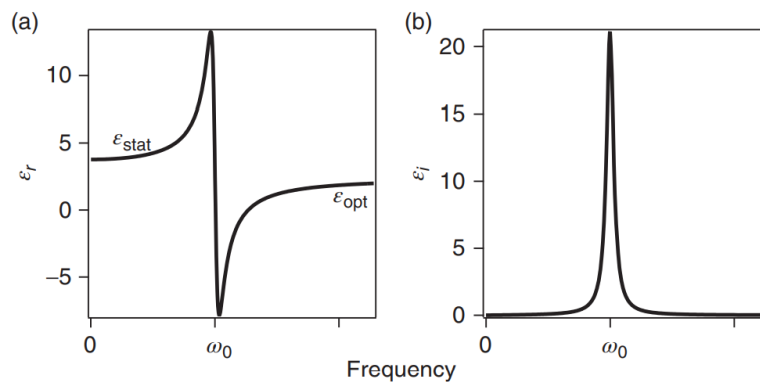
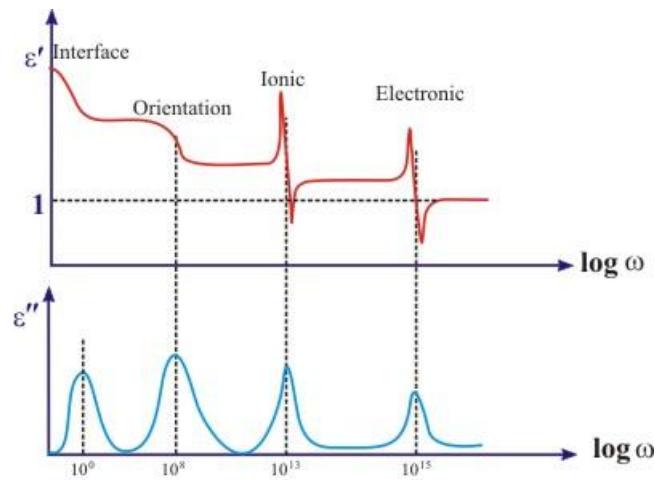


Figure 9.4 Dielectric function for the damped, driven harmonic oscillator close to the resonance frequency ω_0 . (a) Real part and (b) imaginary part of ϵ . ϵ_{stat} and ϵ_{opt} stand for the static and optical value of the real part of ϵ , respectively.



- 4) What is the physical meaning of the imaginary part of the dielectric function?

The imaginary part is a measure of power dissipated by the system and is given by

$$\bar{p} = \frac{1}{2} \epsilon_0 \epsilon_l \omega \mathcal{E}_0^2$$

Problems 9

- 2) *Real crystal structures*: Show that the packing of spheres in a simple cubic lattice fills 52% of the available space.

The length of the cube is a . The radius of the spheres can at maximum be $\frac{a}{2}$ before they overlap

There is a ball at each corner where a quarter of its volume is inside the cube. So the overall volume inside the cube is one sphere with radius $\frac{a}{2}$. We calculate the volume of a sphere

$$V_{ball} = \frac{4}{3}\pi \left(\frac{a}{2}\right)^3 = \frac{\pi a^3}{6}$$

The volume of the cube is

$$V_{space} = a^3$$

The ratio of spheres to space is

$$\frac{V_{ball}}{V_{space}} = \frac{\pi}{6} \approx 0,5235988 \approx 52\%$$

- 3) *Real crystal structures*: Figure 1.16 shows the structures of a two-dimensional hexagonal packed layer of atoms, a hcp crystal, a two-dimensional sheet of carbon atoms arranged in a honeycomb lattice (graphene),

and three-dimensional graphite. (a) Draw a choice of vectors spanning the Bravais lattice for the hexagonal layer of atoms and for graphene, and compare them to each other. (b) Show that the basis for the hexagonal layer contains one atom, while the bases for graphene and the three-dimensional hcp crystal contain two atoms. (c)* Choose lattice vectors for the Bravais lattice of graphite and show that the basis contains four atoms.

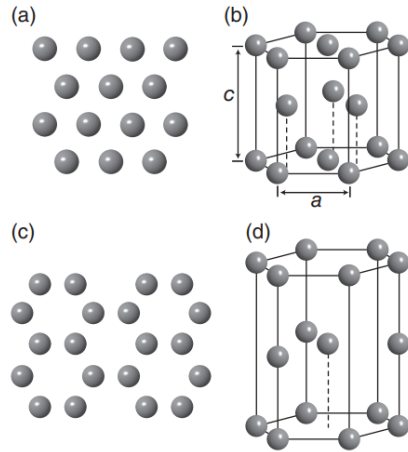
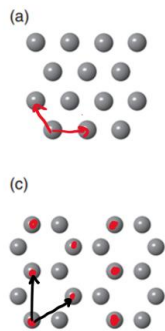


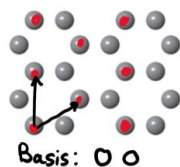
Figure 1.16: (a) Two-dimensional crystal structure of a hexagonal close-packed layer of atoms. (b) Crystal structure of a three-dimensional hcp crystal. (c) Two-dimensional crystal structure of graphene. (d) Three-dimensional crystal structure of graphite (strongly compressed along the c direction). The lines are a mere guide to the eye, not indicating bonds or the size of the unit cell.

a) For the hexagonal layer and graphene

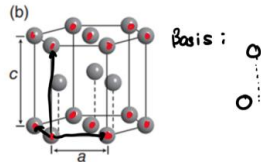


b) Hexagonal layer has basis of 1 because the vectors can reach all of the atoms

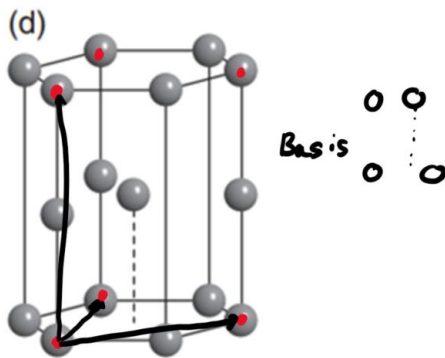
Graphene has a basis of two atom that are adjacent to each other



The three dimensional hcp can be described as



c) Graphite has a basis of 4 atoms



5) *X-ray diffraction*: (a) Determine the maximum wavelength for which constructive interference can be observed in the Bragg model for a simple cubic crystal with a lattice constant of $3.6 \text{ \AA}$. (b) What is the energy of the X-rays in electron volts? (c) If you were to perform neutron diffraction, what kinetic energy would the neutrons need to possess for their de Broglie wavelength to be same? (d) One could argue that if one would use X-rays with twice the wavelength, one would still get a Bragg peak because then constructive interference would occur between the X-rays reflected from every other plane. Why is this argument not valid? (e) One could describe the same crystal by using a unit cell that is a bigger cube of twice the side length, containing eight atoms instead of one. The lattice constant would then be $7.2 \text{ \AA}$. Discuss how this different description would affect the X-ray diffraction from the crystal.

a) The maximum wavelength where constructive interference is observed is given by the condition from Bragg's law

$$2d \sin(\theta) = n\lambda \Rightarrow 2d > \lambda$$

The largest d-spacing is the set of planes $\{100\}$ plane with size $3.6 \text{ \AA}$. Therefore

$$\lambda < 2 \cdot 3.6 \text{ \AA} = 7.2 \text{ \AA}$$

b) The energy of the x-rays is

$$E = hf = \frac{hc}{\lambda} = \frac{6.626 \cdot 10^{-34} J \cdot s \cdot 2.998 \cdot 10^8 \frac{m}{s}}{7.2 \cdot 10^{-10} m \cdot 1.602 \cdot 10^{-19} C} \approx 1722,218 eV$$

c) Neutron

The de Broglie relation is given by

$$\lambda = \frac{h}{p} \Leftrightarrow p = \frac{h}{\lambda}$$

The kinetic energy is given by

$$K = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

Therefore

$$K = \frac{h^2}{2m\lambda^2} = \frac{(6.626 \cdot 10^{-34} J \cdot s)^2}{2 \cdot 1.675 \cdot 10^{-27} kg \cdot (7.2 \cdot 10^{-10} m)^2} \approx 2,528093 \cdot 10^{-22} J$$

In eV

$$K = \frac{2,528093 \cdot 10^{-22} J}{1.602 \cdot 10^{-19} C} \approx 0,001578086 eV$$

- d) When plane 1, 3, 5... constructively interfere then plane 2, 4, 6... will be out of phase and destructively interfere. Therefore we would get zero signal.
- e) The x-ray diffraction would be unaffected because the smallest unit cell would still be the same but using a unit cell that is too large would also mean that you would be missing some reflections in the low 2θ range when you do the XRD measurements.

6) *The reciprocal lattice:* Using the explicit definition of the reciprocal lattice in Eq. (1.16), show first that Eq. (1.17) is obeyed and then proceed, using this relation, to show that the reciprocal lattice defined by Eq. (1.16) does indeed meet the condition from Eq. (1.13).

We have to show using

$$\vec{b}_1 = 2\pi \frac{\vec{a}_2 \times \vec{a}_3}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)}, \quad \vec{b}_2 = 2\pi \frac{\vec{a}_3 \times \vec{a}_1}{\vec{a}_2 \cdot (\vec{a}_3 \times \vec{a}_1)}, \quad \vec{b}_3 = 2\pi \frac{\vec{a}_1 \times \vec{a}_2}{\vec{a}_3 \cdot (\vec{a}_1 \times \vec{a}_2)}$$

That $\vec{a}_i \cdot \vec{b}_j = 2\pi\delta_{ij}$ is obeyed. Before I start doing a lot of calculations let me clarify that the result of a cross product will be perpendicular to the two vectors participating in the cross product. Therefore I can generally say that due to the principle of orthogonality that

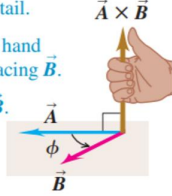
$$\vec{a} \cdot (\vec{a} \times \vec{b}) = \vec{a} \cdot (\vec{b} \times \vec{a}) = 0$$

Also from the Electrodynamics course: $\vec{A} \cdot (\vec{B} \times \vec{C}) = (\vec{A} \times \vec{B}) \cdot \vec{C}$

Also remember that dot products are commutative meaning that $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$

(a) Using the right-hand rule to find the direction of $\vec{A} \times \vec{B}$

- ① Place $\vec{A}$ and $\vec{B}$ tail to tail.
- ② Point fingers of right hand along $\vec{A}$, with palm facing $\vec{B}$.
- ③ Curl fingers toward $\vec{B}$.
- ④ Thumb points in direction of $\vec{A} \times \vec{B}$.



I will let the math speak for itself now

$$\vec{a}_1 \cdot \vec{b}_1 = 2\pi \frac{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)} = 2\pi$$

$$\vec{a}_1 \cdot \vec{b}_2 = 2\pi \frac{\vec{a}_1 \cdot (\vec{a}_3 \times \vec{a}_1)}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)} = 0$$

$$\vec{a}_1 \cdot \vec{b}_3 = 2\pi \frac{\vec{a}_1 \cdot (\vec{a}_1 \times \vec{a}_2)}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)} = 0$$

$$\vec{a}_2 \cdot \vec{b}_1 = 2\pi \frac{\vec{a}_2 \cdot (\vec{a}_2 \times \vec{a}_3)}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)} = 0$$

$$\vec{a}_2 \cdot \vec{b}_2 = 2\pi \frac{\vec{a}_2 \cdot (\vec{a}_3 \times \vec{a}_1)}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)} = 2\pi \frac{(\vec{a}_2 \times \vec{a}_3) \cdot \vec{a}_1}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)} = 2\pi$$

$$\vec{a}_2 \cdot \vec{b}_3 = 2\pi \frac{\vec{a}_2 \cdot (\vec{a}_1 \times \vec{a}_2)}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)} = 0$$

$$\vec{a}_3 \cdot \vec{b}_1 = 2\pi \frac{\vec{a}_3 \cdot (\vec{a}_2 \times \vec{a}_3)}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)} = 0$$

$$\vec{a}_3 \cdot \vec{b}_2 = 2\pi \frac{\vec{a}_3 \cdot (\vec{a}_3 \times \vec{a}_1)}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)} = 0$$

$$\vec{a}_3 \cdot \vec{b}_3 = 2\pi \frac{\vec{a}_3 \cdot (\vec{a}_1 \times \vec{a}_2)}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)} = 2\pi \frac{(\vec{a}_3 \times \vec{a}_1) \cdot \vec{a}_2}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)} = 2\pi \frac{\vec{a}_2 \cdot (\vec{a}_3 \times \vec{a}_1)}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)} = 2\pi \frac{(\vec{a}_2 \times \vec{a}_3) \cdot \vec{a}_1}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)} = 2\pi$$

We have now shown that $\vec{a}_i \cdot \vec{b}_j = 2\pi\delta_{ij}$

We now have to show that $\vec{a}_i \cdot \vec{b}_j = 2\pi\delta_{ij}$ will fulfill the condition $\vec{G} \cdot \vec{R} = 2\pi l$

$$\begin{aligned} \vec{G} \cdot \vec{R} &= (m\vec{b}_1 + n\vec{b}_2 + o\vec{b}_3) \cdot (m'\vec{a}_1 + n'\vec{a}_2 + o'\vec{a}_3) \\ &= m\vec{b}_1 \cdot m'\vec{a}_1 + m\vec{b}_1 \cdot n'\vec{a}_2 + m\vec{b}_1 \cdot o'\vec{a}_3 + n\vec{b}_2 \cdot m'\vec{a}_1 + n\vec{b}_2 \cdot n'\vec{a}_2 + n\vec{b}_2 \cdot o'\vec{a}_3 + o\vec{b}_3 \cdot m'\vec{a}_1 \\ &\quad + o\vec{b}_3 \cdot n'\vec{a}_2 + o\vec{b}_3 \cdot o'\vec{a}_3 \\ &= 2\pi(mm' + nn' + oo') \end{aligned}$$

Let $l = (mm' + nn' + oo')$ then we have shown what we were supposed to

Problems 10

7) *The reciprocal lattice:* For a two-dimensional Bravais lattice

$$\mathbf{R} = m\mathbf{a}_1 + n\mathbf{a}_2, \quad (1.31)$$

the reciprocal lattice is also two-dimensional:

$$\mathbf{G} = m'\mathbf{b}_1 + n'\mathbf{b}_2. \quad (1.32)$$

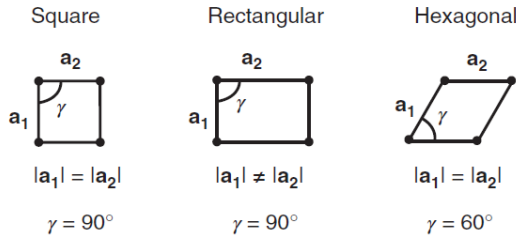


Figure 1.17: Two-dimensional Bravais lattices.

Often, the most practical way to construct the reciprocal lattice is to use the relation

$$\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi \delta_{ij}, \quad (1.33)$$

which remains valid in the two-dimensional case. Find the reciprocal lattices for the three cases displayed in Figure 1.17.

We know that

$$\vec{a}_i \cdot \vec{b}_j = 2\pi \delta_{ij}$$

For the rectangle we define our real lattice vectors as

$$\vec{a}_1 = \begin{pmatrix} 0 \\ |\vec{a}_1| \end{pmatrix}, \quad \vec{a}_2 = \begin{pmatrix} |\vec{a}_2| \\ 0 \end{pmatrix}$$

Therefore

$$\vec{b}_1 = \begin{pmatrix} 0 \\ \frac{2\pi}{|\vec{a}_1|} \end{pmatrix}, \quad \vec{b}_2 = \begin{pmatrix} \frac{2\pi}{|\vec{a}_2|} \\ 0 \end{pmatrix}$$

The square case is the same as the rectangular case but where $|\vec{a}_1| = |\vec{a}_2|$

For the hexagonal case we define the vector as

$$\vec{a}_1 = |\vec{a}_1| \begin{pmatrix} \cos(60) \\ \sin(60) \end{pmatrix} = \frac{|\vec{a}_1|}{2} \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix}$$

$$\vec{a}_2 = \begin{pmatrix} |\vec{a}_2| \\ 0 \end{pmatrix}$$

To calculate the first reciprocal lattice vector we can set up two equations with two unknowns

$$\vec{a}_2 \cdot \vec{b}_1 = \begin{pmatrix} |\vec{a}_2| \\ 0 \end{pmatrix} \cdot \begin{pmatrix} b_{1x} \\ b_{1y} \end{pmatrix} = |\vec{a}_2| b_{1x} = 0$$

$$\Updownarrow$$

$$b_{1x} = 0$$

$$\vec{a}_1 \cdot \vec{b}_1 = \frac{|\vec{a}_1|}{2} \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \cdot \begin{pmatrix} b_{1x} \\ b_{1y} \end{pmatrix} = \frac{|\vec{a}_1|}{2} (b_{1x} + \sqrt{3} b_{1y}) = 2\pi$$

$$\Updownarrow$$

$$\frac{|\vec{a}_1|}{2} \sqrt{3} b_{1y} = 2\pi$$

$$\Updownarrow$$

$$b_{1y} = \frac{4\pi}{\sqrt{3}|\vec{a}_1|}$$

Therefore

$$\vec{b}_1 = \begin{pmatrix} 0 \\ \frac{4\pi}{\sqrt{3}|\vec{a}_1|} \end{pmatrix}$$

For the second vector we do the same thing

$$\vec{a}_2 \cdot \vec{b}_2 = \begin{pmatrix} |\vec{a}_2| \\ 0 \end{pmatrix} \cdot \begin{pmatrix} b_{2x} \\ b_{2y} \end{pmatrix} = |\vec{a}_2| b_{2x} = 2\pi$$

$$\Updownarrow$$

$$b_{2x} = \frac{2\pi}{|\vec{a}_2|}$$

$$\vec{a}_1 \cdot \vec{b}_2 = \frac{|\vec{a}_1|}{2} (b_{2x} + \sqrt{3} b_{2y}) = 0$$

$$\Leftrightarrow$$

$$\frac{|\vec{a}_1|}{2} \frac{2\pi}{|\vec{a}_2|} + \frac{|\vec{a}_1|}{2} \sqrt{3} b_{2x} = 0$$

$$\Leftrightarrow$$

$$\frac{|\vec{a}_1|}{2} \sqrt{3} b_{2x} = -\pi$$

$$\Leftrightarrow$$

$$b_{2x} = -\frac{2\pi}{\sqrt{3}|\vec{a}_1|}$$

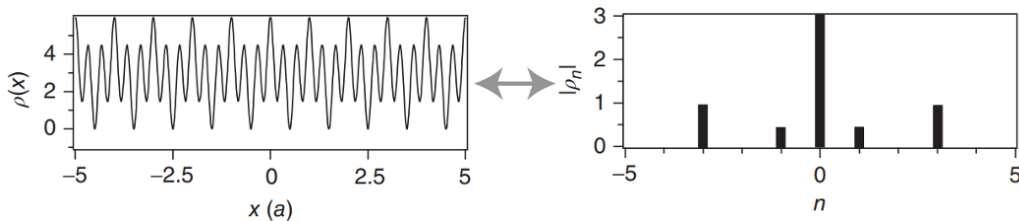
Therefore

$$\vec{b}_1 = \begin{pmatrix} 0 \\ \frac{4\pi}{\sqrt{3}|a_1|} \end{pmatrix}$$

$$\vec{b}_2 = \frac{2\pi}{|a_1|} \begin{pmatrix} 1 \\ -\frac{1}{\sqrt{3}} \end{pmatrix}$$

- 8) *Lattice-periodic functions*: Figure 1.11 shows two lattice-periodic functions and the Fourier coefficients generating these functions. The functions $\rho(x)$ have been calculated from the coefficients ρ_n using Eq. (1.20). (a) Write down the specific Fourier series corresponding to the lower half of Fig. 1.11 and make sure that the result is correct by plotting $\rho(x)$ in the same interval as in the figure, from $-5a$ to $5a$ (you may choose $a = 1$). (b) Now try the reverse operation by numerically calculating the Fourier transform of the obtained $\rho(x)$. Display the resulting $|\rho(k)|$. You will find that it does not match the right-hand side of Fig. 1.11: Instead of δ functions at the reciprocal lattice points, you will obtain quite broad peaks. Why is this so? What would be needed to narrow these peaks? Demonstrate how this can be achieved. (c)* Discuss the implications for the observed intensity in X-ray diffraction. To this end, note that upon fulfilling the Laue condition, the diffracted intensity around a reciprocal lattice vector as in Eq. (1.26) corresponds to the squared norm of the scattering density's Fourier transformation.

a) We need to reproduce this function

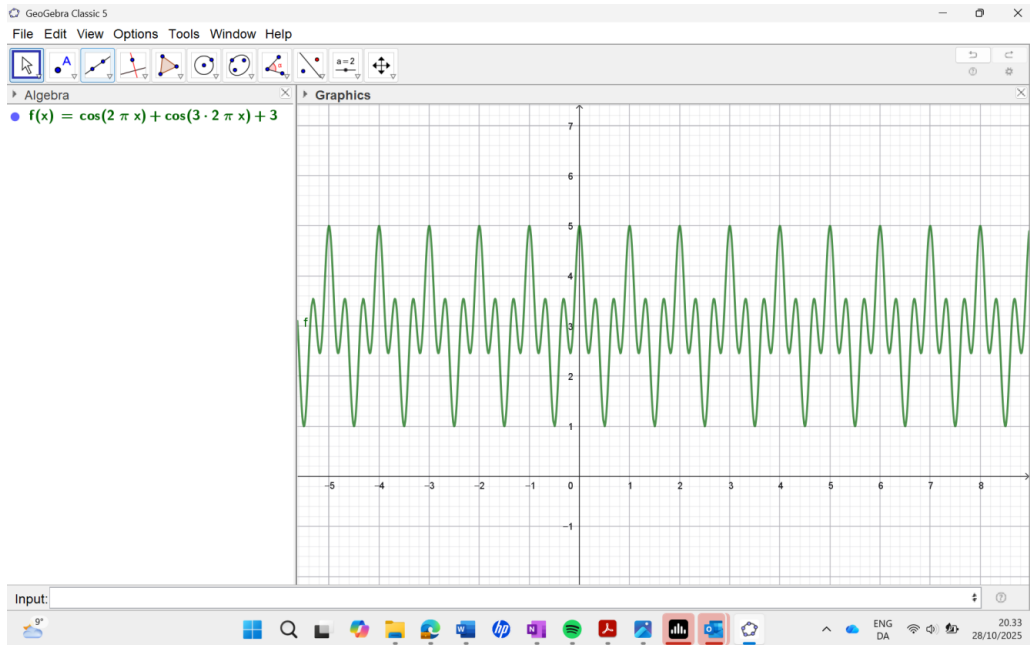


You can kinda guess what the function should be from the Fourier transform. So reading off from the right-hand figure remembering that $\rho_{-n}^* = \rho_n$ such that the bars on the negative n -side adds to the positive n -side

$$\rho(x) = \cos\left(\frac{2\pi}{a}x\right) + 2 \cos\left(3 \cdot \frac{2\pi}{a}x\right) + 3$$

$$\rho(x) = \cos\left(\frac{2\pi}{a}x\right) + 2 \cos\left(\frac{6\pi}{a}x\right) + 3$$

I'm choosing $a = 1$ in my plot and I get exactly the same figure



b) Now we have to calculate the Fourier transform of our function. The Fourier transform is given by

$$\hat{f}(\lambda) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-i\lambda t} dt$$

Now we just calculate

$$\begin{aligned} \hat{f}(\lambda) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\cos\left(\frac{2\pi}{a}t\right) + 2 \cos\left(\frac{6\pi}{a}t\right) + 3 \right) e^{-i\lambda t} dt \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\cos\left(\frac{2\pi}{a}t\right) e^{-i\lambda t} + 2 \cos\left(\frac{6\pi}{a}t\right) e^{-i\lambda t} + 3e^{-i\lambda t} \right) dt \end{aligned}$$

Using the identity $\cos(\theta) = \frac{e^{i\theta} + e^{-i\theta}}{2}$ then we get

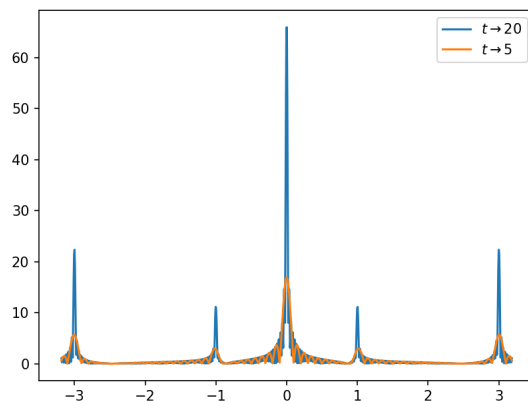
$$\begin{aligned} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\left(\frac{e^{i\frac{2\pi}{a}t} + e^{-i\frac{2\pi}{a}t}}{2} \right) e^{-i\lambda t} + 2 \left(\frac{e^{i\frac{6\pi}{a}t} + e^{-i\frac{6\pi}{a}t}}{2} \right) e^{-i\lambda t} + 3e^{-i\lambda t} \right) dt \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\frac{e^{it(\frac{2\pi}{a}-\lambda)}}{2} + \frac{e^{-it(\frac{2\pi}{a}+\lambda)}}{2} + e^{it(\frac{6\pi}{a}-\lambda)} + e^{-it(\frac{6\pi}{a}+\lambda)} + 3e^{-i\lambda t} \right) dt \end{aligned}$$

$$= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{it(\frac{2\pi}{a}-\lambda)}}{2i(\frac{2\pi}{a}-\lambda)} - \frac{e^{-it(\frac{2\pi}{a}+\lambda)}}{2i(\frac{2\pi}{a}+\lambda)} + \frac{e^{it(\frac{6\pi}{a}-\lambda)}}{i(\frac{6\pi}{a}-\lambda)} - \frac{e^{-it(\frac{6\pi}{a}+\lambda)}}{i(\frac{6\pi}{a}+\lambda)} - \frac{3e^{-i\lambda t}}{i\lambda} \right]_{-\infty}^{\infty}$$

I'm sure if you calculate it analytically then you should be getting some delta functions. However, I have to admit I don't know how to do it unless I spend some extra hours studying for it, so I have calculated it numerically as we were supposed to

$$f(x) = \lim_{k \rightarrow \infty} \frac{1}{\sqrt{2\pi}} \left[\frac{e^{it(\frac{2\pi}{a}-\lambda)}}{2i(\frac{2\pi}{a}-\lambda)} - \frac{e^{-it(\frac{2\pi}{a}+\lambda)}}{2i(\frac{2\pi}{a}+\lambda)} + \frac{e^{it(\frac{6\pi}{a}-\lambda)}}{i(\frac{6\pi}{a}-\lambda)} - \frac{e^{-it(\frac{6\pi}{a}+\lambda)}}{i(\frac{6\pi}{a}+\lambda)} - \frac{3e^{-i\lambda t}}{i\lambda} \right]_{-k}^k$$

The larger the window we calculate the sharper the peaks become. You could also skip all these calculations and use the FFT modules already available in the Scipy package in Python



```
import numpy as np
import matplotlib.pyplot as plt

a=1

def integral(x, t):
    k1 = (2*np.pi/a - x)*1j
    k2 = -(2*np.pi/a + x)*1j
    k3 = (6*np.pi/a - x)*1j
    k4 = -(6*np.pi/a + x)*1j

    f = 1/(np.sqrt(np.pi))*(np.exp(k1*t)/(2*k1)+
                             np.exp(k2*t)/(2*k2)+
                             np.exp(k3*t)/k3+
                             np.exp(k4*t)/k4-
                             3*np.exp(-x*t*1j)/(x*1j)
```

```

    )
    return f

fig, ax = plt.subplots()

xs = np.linspace(-20, 20, 1000)
ys = lambda t: np.abs(integral(xs, t) - integral(xs, -t))

ax.plot(xs/(2*np.pi), ys(20))
ax.plot(xs/(2*np.pi), ys(5))
ax.legend([f'$t \rightarrow 20$', f'$t \rightarrow 5$'])
plt.show()

```

- c) The implications in the XRD experiment is that we observe sharp peaks because the scattering vector has to equal the reciprocal lattice vector. Therefore, we directly image the reciprocal lattice vector and therefore the Fourier transform of our lattice. We do not see peaks of infinite intensity because of imperfections in the experiment e.g. bandwidth of the X-ray source, temperature effects, size effects of our crystal, strain effects and so on.

We also see directly how we can measure the size of any crystal you are measuring with XRD. If your crystals are small then the peaks will be broad whereas sharp peaks means that your crystals are very large.

- 2) *Ionic bonding*: The Madelung constant for a three-dimensional NaCl crystal was given in Section 2.2. (a) Derive the Madelung constant analytically for a one-dimensional NaCl chain as the one shown in Figure 2.5. (b)* Calculate the Madelung constant numerically for a one-dimensional, two-dimensional, and three-dimensional NaCl lattice and plot the result as a function of the number of neighbor “shells” included in the computation. Compare the result for the one-dimensional case to the analytical value from part (a).



Figure 2.5: One-dimensional chain of ions.

- a) 1 dimension

$$A = 2 - \frac{2}{2} + \frac{2}{3} - \frac{2}{4} + \frac{2}{5} - \frac{2}{6} + \frac{2}{7} - \frac{2}{8} + \frac{2}{9} - \frac{2}{10} + \frac{2}{11} - \frac{2}{12} + \frac{2}{13} - \frac{2}{14} + \frac{2}{15} - \frac{2}{16} + \frac{2}{17} - \frac{2}{18} + \frac{2}{19}$$

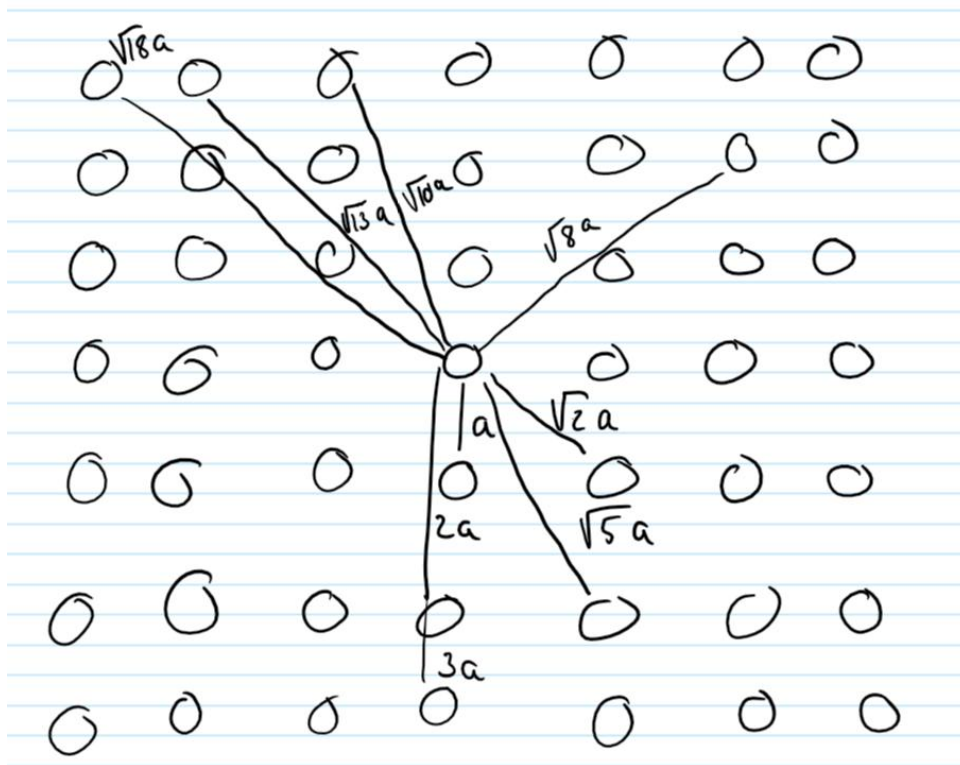
$$= 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 2 \ln(2) = 1,386294361119891$$

b) 2 dimensions

We can find the Madelung constant by using the double sum

$$M = - \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} \frac{(-1)^{n+m}}{\sqrt{n^2 + m^2}}, \quad (n, m) \neq 0$$

Consider that the distance is given by the Pythagorean theorem where the distance from the central atom to an atom at the coordinate (n,m) is $r = \sqrt{n^2 + m^2}$. And for each time n or m changes value then the atoms will change between a negative and positive atom. Hence why the exponent is (n+m). Also we have an extra minus outside of the sum because of definition



Notice that at the corners of each square (x,y) there are 4 equivalent atoms and likewise along the x and y direction. Otherwise there are 8 in each circle.

We can split the calculations into sets of square cells

First square cell

$$A_1 = \left(\frac{4}{1} - \frac{4}{\sqrt{2}} \right) \approx 1,171573$$

Second square cell

$$A_2 = \left(-\frac{4}{2} + \frac{8}{\sqrt{5}} - \frac{4}{\sqrt{8}} \right) \approx 0,1634952$$

Third square cell

$$A_3 = \left(\frac{4}{3} - \frac{8}{\sqrt{10}} + \frac{8}{\sqrt{13}} - \frac{4}{\sqrt{18}} \right) \approx 0,07950295$$

Fourth square cell

$$A_4 = \left(-\frac{4}{4} + \frac{8}{\sqrt{17}} - \frac{8}{\sqrt{20}} + \frac{8}{\sqrt{25}} - \frac{4}{\sqrt{32}} \right) \approx 0,04432384$$

Fifth square cell

$$A_5 = \left(\frac{4}{5} - \frac{8}{\sqrt{26}} + \frac{8}{\sqrt{29}} - \frac{8}{\sqrt{34}} + \frac{8}{\sqrt{41}} - \frac{4}{\sqrt{50}} \right) \approx 0,02834961$$

Sixth square cell

$$A_6 = \left(-\frac{4}{6} + \frac{8}{\sqrt{37}} - \frac{8}{\sqrt{40}} + \frac{8}{\sqrt{45}} - \frac{8}{\sqrt{52}} + \frac{8}{\sqrt{61}} - \frac{4}{\sqrt{72}} \right) \approx 0,01967388$$

Seventh square cell

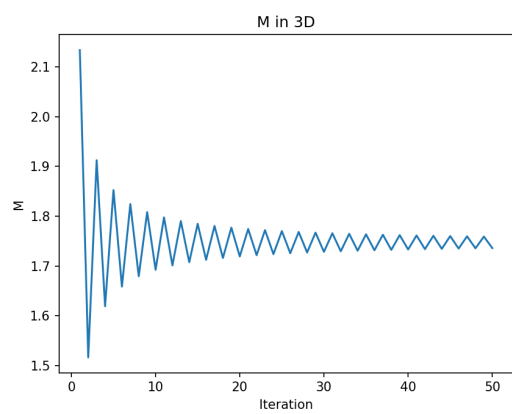
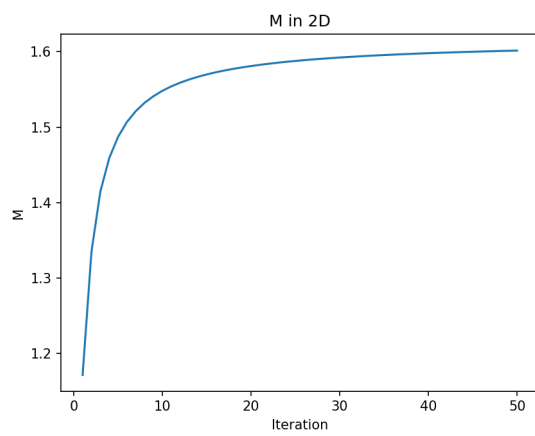
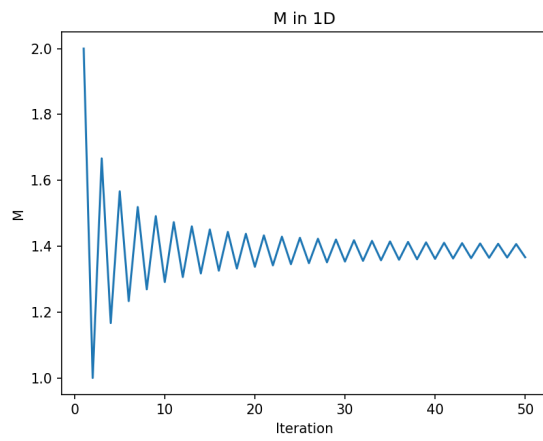
$$A_7 = \left(\frac{4}{7} - \frac{8}{\sqrt{50}} + \frac{8}{\sqrt{53}} - \frac{8}{\sqrt{65}} + \frac{8}{\sqrt{74}} - \frac{8}{\sqrt{85}} + \frac{4}{\sqrt{98}} \right) \approx 0,01298465$$

After all these iterations we get

$$A = 1,171573 + 0,1634952 + 0,07950295 + 0,04432384 + 0,02834961 + 0,01967388 + 0,01298465 \\ \approx 1,519903$$

- c) The 3 dimensional case is an extension of the same logic in the 2 dimensional case. We can write the Madelung constant as

$$M = - \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} \frac{(-1)^{n+m+l}}{\sqrt{n^2 + m^2 + l^2}}, \quad (n, m, l) \neq (0,0,0)$$



Inefficient code that could be optimized but I'm lazy

```
import matplotlib.pyplot as plt
import numpy as np

# 1 dimensional
iterations = 50
index = np.arange(1, iterations+1)
M = []

nmax = 1
def function1d(n):
    return -(-1)**(n)/(np.sqrt(n**2))

for nmax in range(1, iterations+1):
    Mk = []
    for n in range(-nmax, nmax+1):
        if (n==0):
            None
        else:
            Mk.append(function1d(n))
    M.append(np.sum(Mk))

fig, ax = plt.subplots()
ax.plot(index, M)
ax.set_title('M in 1D')
ax.set_xlabel('Iteration')
ax.set_ylabel('M')
plt.show()

# 2 dimensional
iterations = 50
index = np.arange(1, iterations+1)
M = []

nmax = 1
def function2d(n, m):
    return -(-1)**(n+m)/(np.sqrt(n**2+m**2))

for nmax in range(1, iterations+1):
    Mk = []
    for n in range(-nmax, nmax+1):
        for m in range(-nmax, nmax+1):
            if (n==0 and m==0):
```

```

        None
    else:
        Mk.append(function2d(n,m))
    M.append(np.sum(Mk))

fig, ax = plt.subplots()
ax.plot(index, M)
ax.set_title('M in 2D')
ax.set_xlabel('Iteration')
ax.set_ylabel('M')
plt.show()

# 3 dimensional
iterations = 50
index = np.arange(1,iterations+1)
M = []

nmax = 1
def function3d(n, m, l):
    return -(-1)**(n+m+l)/(np.sqrt(n**2+m**2+l**2))

for nmax in range(1, iterations+1):
    Mk = []
    for n in range(-nmax, nmax+1):
        for m in range(-nmax, nmax+1):
            for l in range(-nmax, nmax+1):
                if (n==0 and m==0 and l==0):
                    None
                else:
                    Mk.append(function3d(n,m,l))
    M.append(np.sum(Mk))

fig, ax = plt.subplots()
ax.set_title('M in 3D')
ax.plot(index, M)
ax.set_xlabel('Iteration')
ax.set_ylabel('M')
plt.show()

```

- 3) *Covalent bonding*: (a) Given the linear system of equations (2.7) and the solution in Eq. (2.8), show that for the bonding “+” solution $c_1 = c_2$ and for the antibonding “-” solution $c_1 = -c_2$. (b) We could thus choose the resulting wave functions to be $\Psi(\mathbf{r}) = \phi_A(\mathbf{r}) \pm \phi_B(\mathbf{r})$, but these are not normalized. Calculate the proper normalization factor, assuming that the atomic wave functions $\phi_A(\mathbf{r})$ and $\phi_B(\mathbf{r})$ are normalized.

a) Solutions

$$c_1(H'_{AA} - E') + c_2(H'_{AB} - SE') = 0$$

$$c_1(H'_{AB} - SE') + c_2(H'_{AA} - E') = 0$$

$$E'_{\pm} = \frac{H'_{AA} \pm H'_{AB}}{1 \pm S}$$

Solving the system of equations give

$$c_1 = -\frac{c_2(H'_{AB} - SE')}{H'_{AA} - E'}$$

For the + solution we let $E' = E'_+$

$$\begin{aligned} c_1 &= -\frac{c_2 \left(H'_{AB} - S \left(\frac{H'_{AA} + H'_{AB}}{1 + S} \right) \right)}{H'_{AA} - \left(\frac{H'_{AA} + H'_{AB}}{1 + S} \right)} \\ &= -\frac{c_2 (H'_{AB}(1 + S) - S(H'_{AA} + H'_{AB}))}{H'_{AA}(1 + S) - (H'_{AA} + H'_{AB})} \\ &= -\frac{c_2 (H'_{AB} + H'_{AB}S - SH'_{AA} - SH'_{AB})}{H'_{AA} + H'_{AA}S - H'_{AA} - H'_{AB}} \\ &= -\frac{c_2 (H'_{AB} - SH'_{AA})}{H'_{AA}S - H'_{AB}} \\ &= c_2 \end{aligned}$$

For the - solution we let $E' = E'_-$

$$c_1 = -\frac{c_2 \left(H'_{AB} - S \left(\frac{H'_{AA} - H'_{AB}}{1 - S} \right) \right)}{H'_{AA} - \left(\frac{H'_{AA} - H'_{AB}}{1 - S} \right)}$$

$$\begin{aligned}
&= -\frac{c_2(H'_{AB}(1-S) - S(H'_{AA} - H'_{AB}))}{H'_{AA}(1-S) - (H'_{AA} - H'_{AB})} \\
&= -\frac{c_2(H'_{AB} - H'_{AB}S - SH'_{AA} + SH'_{AB})}{H'_{AA} - H'_{AA}S - H'_{AA} + H'_{AB}} \\
&= -\frac{c_2(H'_{AB} - SH'_{AA})}{-H'_{AA}S + H'_{AB}} \\
&= -c_2
\end{aligned}$$

This is exactly what we wanted to show

Mortens løsning. Antag $c_1=c_2$ og isoler for E i den ene ligning og se at det giver plus løsningen. Gentag for $c_1=-c_2$

b) We now define a wavefunction under the assumption that ϕ_A and ϕ_B are already normalized

$$\psi_{\pm}(\vec{r}) = N_{\pm}(\phi_A(\vec{r}) \pm \phi_B(\vec{r}))$$

We can now calculate the normalization factor

$$1 = \langle \psi | \psi \rangle = \langle N_{\pm}(\phi_A \pm \phi_B) | N_{\pm}(\phi_A \pm \phi_B) \rangle = N_{\pm}^2 (\langle \phi_A | \phi_A \rangle \pm \langle \phi_A | \phi_B \rangle \pm \langle \phi_B | \phi_A \rangle + \langle \phi_B | \phi_B \rangle)$$

$$N_{\pm}^2 = \frac{1}{2 \pm 2\langle \phi_A | \phi_B \rangle}$$

$$N_{\pm} = \frac{1}{\sqrt{2 \pm 2S}}$$

Where $S = \langle \phi_A | \phi_B \rangle$

4) *Metallic bonding*: The most important contribution to the stability gained by metallic bonding is the lowering of kinetic energy. To verify this, consider an electron in a one-dimensional box, inside of which the

potential is zero and outside of which the potential is infinite. Consider first a box with a length corresponding to the size of an atom – say, twice the Bohr radius – and calculate the lowest energy eigenvalue (you can consider any standard text in quantum mechanics for help with this problem). Give the result in electron volts. Clearly, this energy is only kinetic energy. By how much is the kinetic energy lowered when you increase the size of the box by a factor of 10, so that it is roughly the size of the interatomic spacing in a crystal?

This is simply an infinite box

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2m_e a^2}$$

The lowest energy eigenstate is

$$E_1 = \frac{\pi^2 \hbar^2}{2m_e a^2}$$

The Bohr radius is $0.529 \text{ \AA}$ which gives

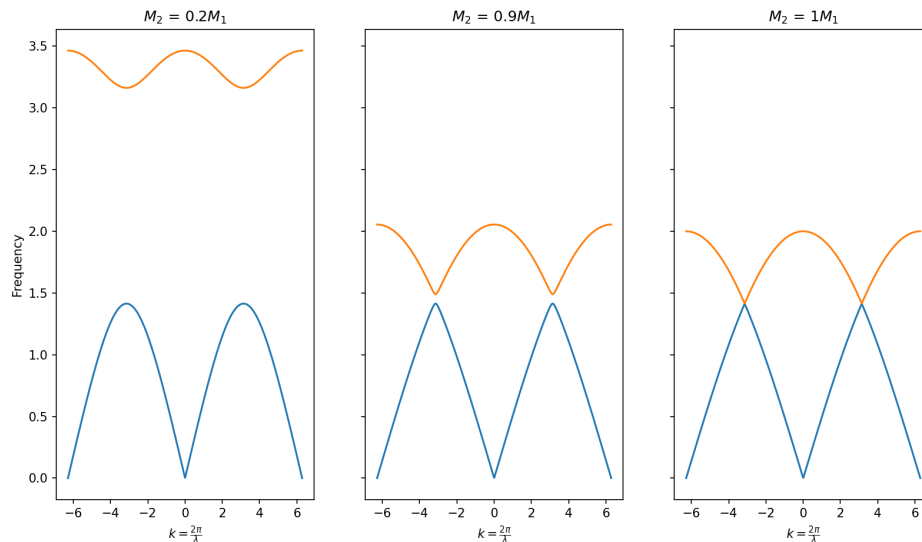
$$E_1 = \frac{\pi^2 \cdot (1.0546 \cdot 10^{-34} \text{ J} \cdot \text{s})^2}{2 \cdot 9.1093837 \cdot 10^{-31} \text{ kg} \cdot (2 \cdot 0.529 \cdot 10^{-10} \text{ m})^2 \cdot 1.6022 \cdot 10^{-19} \text{ C}} = 33,5945 \text{ eV}$$

When the box is increased by a factor 10 then the energy is decreased by a factor 100

Problems 11

- 2) *One-dimensional chain with two atoms per unit cell:* For two atoms per unit cell of length b , we obtain two branches in the phonon dispersion, the acoustic and the optical branch. The solutions are given by Eq. (4.17). (a) Plot these solutions inside the first Brillouin zone for $M_2 = 0.2 M_1$, $M_2 = 0.9 M_1$, and $M_2 = M_1$. (b) Consider the case where $M_1 \neq M_2$. For longitudinal vibrations, sketch how the atoms would move for k close to 0 and for k at the Brillouin zone boundary ($k = \pi/b$). (c) Which movement corresponds to which solution of Eq. (4.17)? Hint: This can be determined by considering the relative amplitudes for the two atoms, using e.g. Eq. (4.15). (d) Explain what happens in the case of $M_2 = M_1$ discussed in (a).

a) This is just pure Python. Here is a quick and dirty script



```
import numpy as np
import matplotlib.pyplot as plt

b = 1
gamma = 1

M1 = 1
M2 = 0.2*M1

def omega(k, sol):
    front_factor = gamma*(1/M1 + 1/M2)
    second_factor = gamma*np.sqrt((1/M1 + 1/M2)**2 - 4/(M1*M2)) *
np.sin(k*b/2)**2
```

```

    if sol == 1:
        return np.sqrt(front_factor + second_factor)

    if sol == 0:
        return np.sqrt(front_factor - second_factor)

fig, axs = plt.subplots(1, 3, sharey=True)

xs = np.linspace(-2*np.pi/b, 2*np.pi/b, 1000)
ys1 = omega(xs, 0)
ys2 = omega(xs, 1)

ax = axs[0]
ax.plot(xs, ys1)
ax.plot(xs, ys2)
ax.set_xlabel('$k=\\frac{2\\pi}{\\lambda}$')
ax.set_ylabel(r'Frequency')
ax.set_title(f'$M_2$ = {M2}$M_1$')

M2 = 0.9*M1
xs = np.linspace(-2*np.pi/b, 2*np.pi/b, 1000)
ys1 = omega(xs, 0)
ys2 = omega(xs, 1)

ax = axs[1]
ax.plot(xs, ys1)
ax.plot(xs, ys2)
ax.set_xlabel('$k=\\frac{2\\pi}{\\lambda}$')
ax.set_title(f'$M_2$ = {M2}$M_1$')

M2 = M1
xs = np.linspace(-2*np.pi/b, 2*np.pi/b, 1000)
ys1 = omega(xs, 0)
ys2 = omega(xs, 1)

ax = axs[2]
ax.plot(xs, ys1)
ax.plot(xs, ys2)
ax.set_xlabel('$k=\\frac{2\\pi}{\\lambda}$')
ax.set_title(f'$M_2$ = {M2}$M_1$')

plt.show()

```

b) + c)

To better understand how the atoms may be moving let us first look at

First we take a look at

$$u_n(t) = ue^{i(kbn-\omega t)}$$

$$v_n(t) = ve^{i(kbn-\omega t)}$$

In the case where $k = 0$ then

$$u_n(t) = u_{n+1}(t) = ue^{-i\omega t}$$

$$v_n(t) = v_{n+1}(t) = ve^{-i\omega t}$$

To figure out how the amplitudes are related we will have to look at

$$(2\gamma - \omega^2 M_1)u - \gamma(e^{-ikb} + 1)v = 0$$

$$-\gamma(e^{ikb} + 1)u + (2\gamma - \omega^2 M_2)v = 0$$

$$\Updownarrow$$

$$\frac{u}{v} = \frac{\gamma(1 + e^{-ikb})}{2\gamma - M_1\omega^2} = \frac{(2\gamma - \omega^2 M_2)}{\gamma(e^{ikb} + 1)}$$

For $k = 0$ we find that

$$\frac{u}{v} = \frac{(2\gamma - \omega^2 M_2)}{2\gamma} = 1 - \frac{\omega^2 M_2}{2\gamma}$$

And ω^2 for $k = 0$ gives

$$\omega^2 = \gamma\left(\frac{1}{M_1} + \frac{1}{M_2}\right) \pm \gamma\left[\left(\frac{1}{M_1} + \frac{1}{M_2}\right)^2 - \frac{4}{M_1 M_2} \sin^2(0)\right]^{\frac{1}{2}}$$

$$\omega^2 = \gamma\left(\frac{1}{M_1} + \frac{1}{M_2}\right) \pm \gamma\left(\frac{1}{M_1} + \frac{1}{M_2}\right) = \begin{cases} 2\gamma\left(\frac{1}{M_1} + \frac{1}{M_2}\right), & \text{optical branch} \\ 0, & \text{acoustic branch} \end{cases}$$

This gives the two solutions

$$\frac{u}{v} = 1 - \frac{\omega^2 M_2}{2\gamma} = \begin{cases} -\frac{M_2}{M_1}, & \text{optical branch} \\ 1, & \text{acoustic branch} \end{cases}$$

We now have a picture of how the the atoms would move in the case where $k = 0$ then

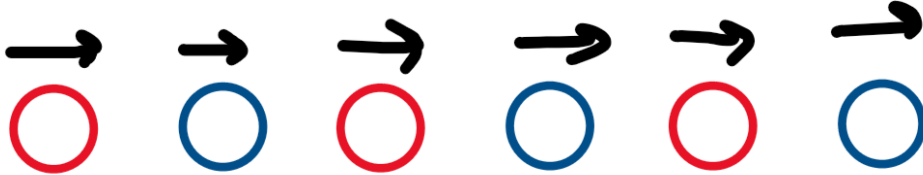
$$u_n(t) = u_{n+1}(t) = ue^{-i\omega t}$$

$$v_n(t) = v_{n+1}(t) = ve^{-i\omega t}$$

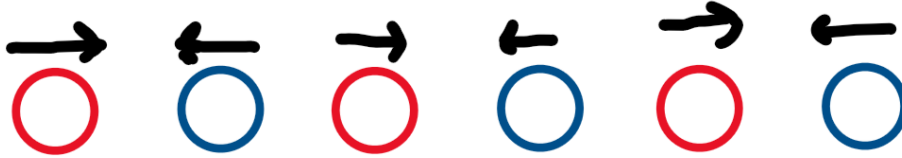
$$\frac{u}{v} = 1 - \frac{\omega^2 M_2}{2\gamma} = \begin{cases} -\frac{M_2}{M_1}, & \text{optical branch} \\ 1, & \text{acoustic branch} \end{cases}$$

$$\omega^2 = \gamma\left(\frac{1}{M_1} + \frac{1}{M_2}\right) \pm \gamma\left(\frac{1}{M_1} + \frac{1}{M_2}\right) = \begin{cases} 2\gamma\left(\frac{1}{M_1} + \frac{1}{M_2}\right), & \text{optical branch} \\ 0, & \text{acoustic branch} \end{cases}$$

In the acoustic branch, the fact that $\omega = 0$ implies that no atoms will be moving is true. But if we are extremely close to $k = 0$ then we still have some vibration where all atoms move in the same direction since $\frac{u}{v} = 1$



In the optical branch all the atoms move in phase but the amplitudes between u and v have opposite signs. Therefore we have the following image where the wavelength is practically infinite



Now we have to repeat this for $k = \frac{\pi}{b}$ and again we start by looking at

$$u_n(t) = u e^{i(kbn - \omega t)}$$

$$v_n(t) = v e^{i(kbn - \omega t)}$$

We see that for $k = \frac{\pi}{b}$

$$u_n(t) = u e^{i(\pi n - \omega t)} = -u e^{i(\pi(n+1) - \omega t)} = -u_{n+1}(t)$$

$$v_n(t) = -v_{n+1}(t)$$

Now what about the amplitudes

$$\frac{u}{v} = \frac{\gamma(1 + e^{-ikb})}{2\gamma - M_1\omega^2} = \frac{(2\gamma - \omega^2 M_2)}{\gamma(e^{ikb} + 1)}$$

If we insert $k = \frac{\pi}{b}$, it will be confusing here because it implies that both $\frac{u}{v} = 0$ and $\frac{u}{v} \rightarrow \infty$. To solve this mystery let us try to calculate the frequency first

$$\begin{aligned} \omega^2 &= \gamma \left(\frac{1}{M_1} + \frac{1}{M_2} \right) \pm \gamma \left[\left(\frac{1}{M_1} + \frac{1}{M_2} \right)^2 - \frac{4}{M_1 M_2} \sin^2 \left(\frac{\pi}{2} \right) \right]^{\frac{1}{2}} \\ &= \gamma \left(\frac{1}{M_1} + \frac{1}{M_2} \right) \pm \gamma \left[\left(\frac{1}{M_1} + \frac{1}{M_2} \right)^2 - \frac{4}{M_1 M_2} \right]^{\frac{1}{2}} \end{aligned}$$

$$\begin{aligned}
&= \gamma \left(\frac{1}{M_1} + \frac{1}{M_2} \right) \pm \gamma \left[\frac{1}{M_1^2} + \frac{1}{M_2^2} + \frac{2}{M_1 M_2} - \frac{4}{M_1 M_2} \right]^{\frac{1}{2}} \\
&= \gamma \left(\frac{1}{M_1} + \frac{1}{M_2} \right) \pm \gamma \left[\frac{1}{M_1^2} + \frac{1}{M_2^2} - \frac{2}{M_1 M_2} \right]^{\frac{1}{2}} \\
&= \gamma \left(\frac{1}{M_1} + \frac{1}{M_2} \right) \pm \gamma \left[\left(\frac{1}{M_1} - \frac{1}{M_2} \right)^2 \right]^{\frac{1}{2}} \\
&= \gamma \left(\frac{1}{M_1} + \frac{1}{M_2} \right) \pm \gamma \left(\frac{1}{M_1} - \frac{1}{M_2} \right) \\
&= \begin{cases} \frac{2\gamma}{M_1} \\ \frac{2\gamma}{M_2} \end{cases}
\end{aligned}$$

Now whether which solution is the optical branch and the acoustic branch depends on which mass is larger. The optical branch always has the higher frequency than the acoustic branch. This is just a matter of definition where we without loss of generality can claim that $M_1 > M_2$ then

$$\omega^2 = \begin{cases} \frac{2\gamma}{M_1}, & \text{acoustic branch} \\ \frac{2\gamma}{M_2}, & \text{optical branch} \end{cases}$$

Let us insert this into

$$\frac{u}{v} = \frac{\gamma(1 + e^{-ikb})}{2\gamma - M_1 \omega^2} = \frac{(2\gamma - \omega^2 M_2)}{\gamma(e^{ikb} + 1)}$$

We then find that for the acoustic branch that we cannot use one solution because we get the following

$$\frac{u}{v} = \frac{\gamma(1 + e^{-ikb})}{2\gamma - M_1 \omega^2} = \frac{0}{0} \rightarrow \text{Undefined}$$

But if we use the other equation with $k = \frac{\pi}{b}$

$$\frac{u}{v} = \frac{(2\gamma - \omega^2 M_2)}{\gamma(e^{ikb} + 1)} \Rightarrow v = 0$$

We then find that for the optical branch that we cannot use one solution because we get the following

$$\frac{u}{v} = \frac{(2\gamma - \omega^2 M_2)}{\gamma(e^{ikb} + 1)} = \frac{0}{0} \rightarrow \text{Undefined}$$

And

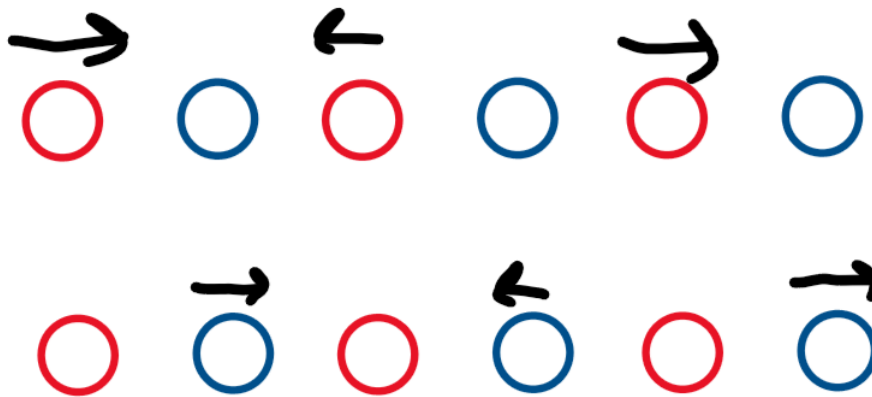
$$\frac{u}{v} = \frac{\gamma(1 + e^{-ikb})}{2\gamma - M_1 \omega^2} = 0$$

Nice now we have a better picture so let us gather the facts

$$u_n(t) = -u_{n+1}(t)$$

$$v_n(t) = -v_{n+1}(t)$$

And $u = 0$ in the optical branch while $v = 0$ in the acoustic branch. This gives us the following pictures



d)

If $M_2 = M_1$ then we reduce the problem to a monoatomic chain. We also see that when we plot the optical branch and acoustic branch together in a)

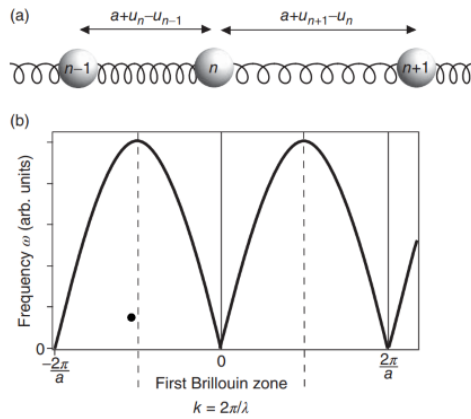


Figure 4.1 (a) One-dimensional chain with one atom per unit cell. (b) Allowed vibrational frequencies ω as a function of the wave vector k .

- 3) *Periodic boundary conditions:* Periodic boundary conditions lead to a restriction of the possible k values. Consider a linear chain of copper atoms. Assume the length of the chain to be 1 cm, the lattice spacing to be 0.36 nm, and the force constant to be 50 N m^{-1} . Calculate the smallest possible finite wave vector k . What is the vibrational angular frequency for this wave vector? What is the corresponding energy in eV and temperature in K?

The smallest wave vector has $m = 1$

$$k_{min} = \frac{2\pi}{aN} = \frac{2\pi}{L} = \frac{2\pi}{1\text{cm}} = 2\pi \text{ cm}^{-1}$$

We can calculate the frequency for this k-vector which is

$$\omega(k) = 2\sqrt{\frac{\gamma}{M}} \left| \sin\left(\frac{ka}{2}\right) \right|$$

Where

$$M = \frac{m}{N_A} = \frac{63.546 \frac{\text{g}}{\text{mol}}}{6.022 \cdot 10^{23} \text{mol}^{-1}}$$

Thus the vibrational frequency is

$$\omega = 2 \sqrt{\frac{50 \cdot \text{N} \cdot \text{m}^{-1}}{63.546 \frac{\text{g}}{\text{mol}}}} \left| \sin\left(\frac{2\pi \text{ cm}^{-1} \cdot 0.36 \text{ nm}}{2}\right) \right| = 4,923723 \cdot 10^6 \text{ s}^{-1}$$

To convert it to energy we use

$$E = hf = \frac{h\omega}{2\pi} = \frac{6.626 \cdot 10^{-34} \text{ J} \cdot \text{s} \cdot 4,923723 \cdot 10^6 \text{ s}^{-1}}{2\pi} = 5,192365 \cdot 10^{-28} \text{ J} \\ \approx 3,2408195975912 \cdot 10^{-9} \text{ eV}$$

To convert it to temperature we use

$$E = kT \\ T = \frac{E}{k} = \frac{5,192365 \cdot 10^{-28} \text{ J}}{1.3806 \cdot 10^{-23} \frac{\text{J}}{\text{K}}} \approx 3,760948 \cdot 10^{-5} \text{ K}$$

- 4) *Vibrations in two dimensions:* Figure 4.17a shows a two-dimensional square lattice with one atom per unit cell, lattice constant a and atoms of mass M that interact with their nearest neighbors via force constants γ . The phonon dispersion as a function of a two-dimensional wave vector $\mathbf{k} = (k_x, k_y)$ is

$$\omega(\mathbf{k}) = \sqrt{\frac{2\gamma}{M}} \sqrt{2 - \cos k_x a - \cos k_y a}. \quad (4.54)$$

Plot this dispersion along one-dimensional cuts in the two-dimensional Brillouin zone shown in Figure 4.17b. Choose a path from Γ ($\mathbf{k} = (0, 0)$) to X ($\mathbf{k} = (-\pi/a, 0)$) to M ($\mathbf{k} = (-\pi/a, \pi/a)$) and back to Γ .

Simply plot

$$\omega(k) = \sqrt{2 - \cos(k_x) - \cos(k_y)}$$

From Γ to X then k_y does not vary so we plot

$$\omega(k) = \sqrt{2 - \cos(k_x) - \cos(0)}, \quad k_x \in [0, -\pi]$$

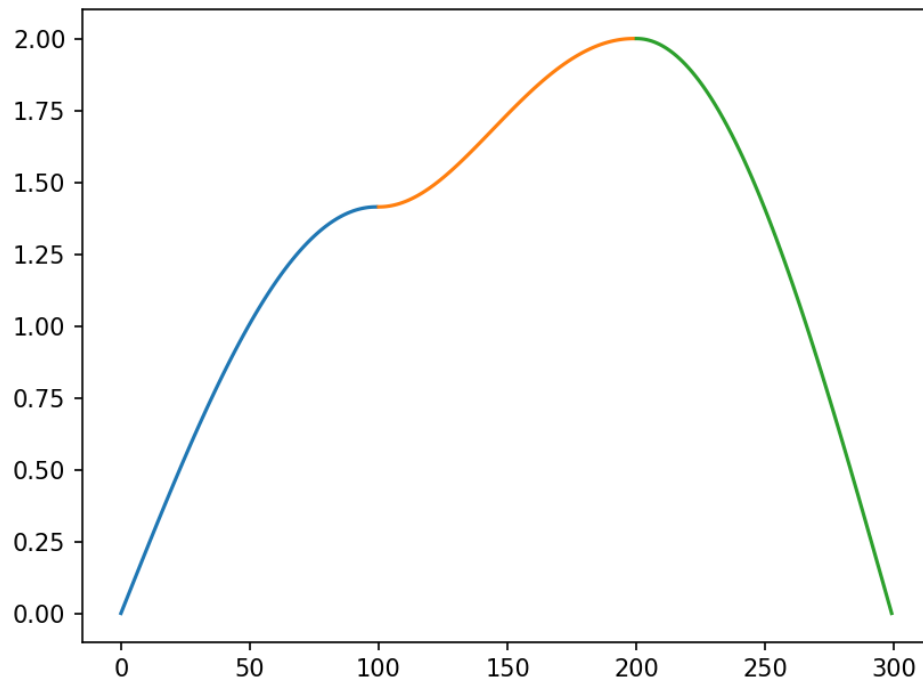
From X to M then k_x does not vary so we let

$$\omega(k) = \sqrt{2 - \cos(-\pi) - \cos(k_y)}, \quad k_y \in [0, \pi]$$

From M to Γ then we let both vary

$$\omega(k) = \sqrt{2 - \cos(k_x) - \cos(k_y)}$$

Where each step size is equal. The result is shown below where each color is each path



```
import numpy as np
import matplotlib.pyplot as plt

def function(x,y):
    return np.sqrt(2-np.cos(x)-np.cos(y))

GX = np.linspace(0, -np.pi, 100)
omega_GX = function(GX, 0)

XM = np.linspace(0, np.pi, 100)
omega_XM = function(-np.pi, XM)

MGx = np.linspace(-np.pi, 0, 100)
MGy = np.linspace(np.pi, 0, 100)
omega_MG = function(MGx, MGy)

fig, ax = plt.subplots()
ax.plot(range(0, len(GX)), omega_GX)
ax.plot(range(len(GX), len(GX)+len(XM)), omega_XM)
ax.plot(range(len(GX)+len(XM), len(GX)+len(XM)+len(MGx)), omega_MG)
plt.show()
```

- 7) *Atomic force constants and Debye temperature:* (a) Estimate the values of the force constant γ and the angular frequency ω for the atomic vibrations in copper. Use the Young's modulus, $Y = 130 \text{ GPa}$, and the cubic lattice constant, 0.36 nm (ignore the fact that copper actually crystallizes in an fcc structure, not in a simple cubic structure). (b) Estimate the angular frequency for the vibrations that correspond to the Debye temperature of 343 K and compare it to the result in (a). (c) Estimate the amplitude of the vibrations as a fraction of the lattice spacing at room temperature.

a) Young's modulus is given by

$$Y = \frac{\sigma}{\varepsilon} = \frac{\gamma}{a}$$

$$\Updownarrow$$

$$\gamma = Ya = 130 \text{ GPa} \cdot 0.36 \text{ nm} = 46,8 \frac{\text{N}}{\text{m}}$$

The highest angular frequency is

$$\omega = \sqrt{\frac{\gamma}{M}} = \sqrt{\frac{46,8 \frac{\text{N}}{\text{m}}}{\frac{63,546 \frac{\text{g}}{\text{mol}}}{6,022 \cdot 10^{23} \text{ mol}^{-1}}}} \approx 2,10596 \cdot 10^{13} \text{ s}^{-1}$$

b) The Debye temperature is given by

$$\Theta_D = \frac{\hbar \omega_D}{k_B}$$

$$\Updownarrow$$

$$\omega_D = \frac{\Theta_D k}{\hbar} = \frac{343 \text{ K} \cdot 1,381 \cdot 10^{-23} \frac{\text{J}}{\text{K}}}{1,055 \cdot 10^{-34} \text{ J} \cdot \text{s}} \approx 4,48989 \cdot 10^{13} \text{ s}^{-1}$$

The angular frequency is larger

c) From the harmonic approximation the amplitude of the vibration is given by

$$x_m = \sqrt{\frac{2kT}{\gamma}} = \sqrt{\frac{2kT}{Ya}}$$

$$= \sqrt{\frac{2 \cdot 1.381 \cdot 10^{-23} \frac{J}{K} \cdot 293 K}{130 GPa \cdot 0.36 nm}} \cdot a = 0,0365 a$$

Problems 12

6) *Bloch electrons*: Show that the spatial probability density for one-dimensional free electrons is constant. (b) Show that it has the periodicity of the corresponding Bravais lattice for Bloch electrons.

a) The wave function for a one-dimensional free electron is

$$\psi(x) = Ae^{-ikx}$$

The spatial probability is calculated as we know from the quantum mechanics course

$$|\psi|^2 = \psi^*\psi = |A|^2 e^{-ikx} e^{ikx} = |A|^2$$

Therefore the spatial probability density is constant since A is a constant

b) This questions sounds a bit weird for me but I assume they mean something along the lines of “Show that the spatial probability density of Bloch electrons has the periodicity of the corresponding Bravais lattice”

From Bloch’s theorem we know that the wavefunction can be described as a plane wave modified by a lattice periodic function

$$\psi_k(r) = u_k(r)e^{ikr}$$

Where u_k is the lattice periodic function

$$u_k(r) = u_k(r + R) = \sum_G c_{k-G} e^{iGr}$$

We do the same thing and find an expression for the lattice periodic function

$$|\psi_k|^2 = |u_k(r)e^{ikr}|^2 = u_k(r)e^{ikr} u_k^*(r)e^{-ikr} = |u_k(r)|^2$$

Now since

$$|\psi_k|^2 = |u_k(r)|^2 = |u_k(r + R)|^2$$

Then we have shown that the spatial probability density has the periodicity of the Bravais lattice

- 7) *Metals and nonmetals:* Consider a one-dimensional chain of N atoms with one atom per unit cell. Assume periodic boundary conditions and that each atom has Z valence electrons. (a) Show that you can fill exactly $Z/2$ bands with these electrons or, equivalently, that each band can accommodate $2N$ electrons. Hint: Consider the number of possible k -points given by the periodic boundary conditions and keep in mind that each state can be occupied by two electrons. (b) Figure 6.13 shows that Si has four filled bands (for some values of k , the energies of the bands are degenerate, but not for all). There are also four electrons per Si atom (not eight!). Explain why this is so. (c) Having an even number of electrons per unit is necessary but not sufficient for a solid to be a semiconductor/insulator. Give an example for an elemental solid that is a metal despite having an even number of electrons per unit cell.

a)

The number of k -points in each band is given by

$$k = \frac{2\pi}{aN} m, \quad m = 1, 2, 3, \dots, N$$

Therefore there are N k -points in each band. Furthermore, each k -point can hold 2 electrons since they can be in spin up or spin down configuration. Therefore, when each atom has Z valence electrons then $Z/2$ bands would be filled

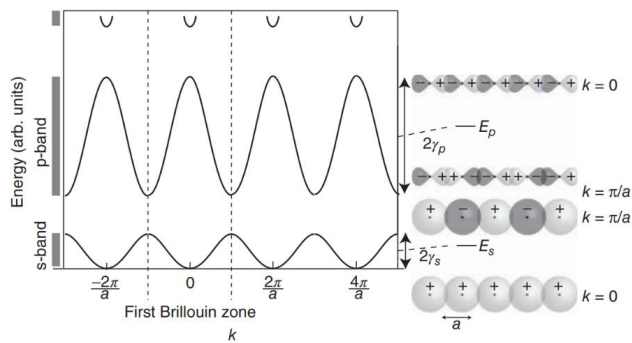
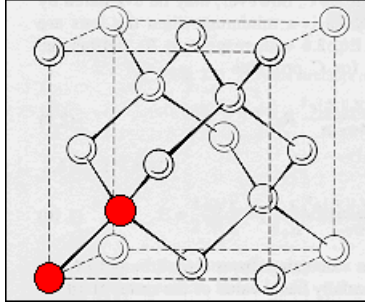


Figure 6.11 Bands for a one-dimensional solid calculated in the tight-binding approximation. The right-hand side also shows the Bloch wave functions for the s - and p -band

for $k = 0$ and $k = \pi/a$. The black dots symbolize the position of the nuclei and the two different shades of gray symbolize the sign of the wave function.

- b) We need to remember that we have to take into account how many atoms are in the primitive unit cell. Si has a basis of two atoms which means that the primitive unit cell also contains two atoms and therefore 8 electrons. One band can accommodate two valence electrons per primitive unit cell which gives us 4 filled bands



- c) Graphene has two atoms per primitive unit cell and each atom has 4 electrons which means there is 8 electrons per primitive unit cell. Hence 4 bands can be filled which are the three σ -bands and one π -band. The π^* -band is unoccupied and the lack of band gap between π and π^* at $k = K$ makes it a semimetal

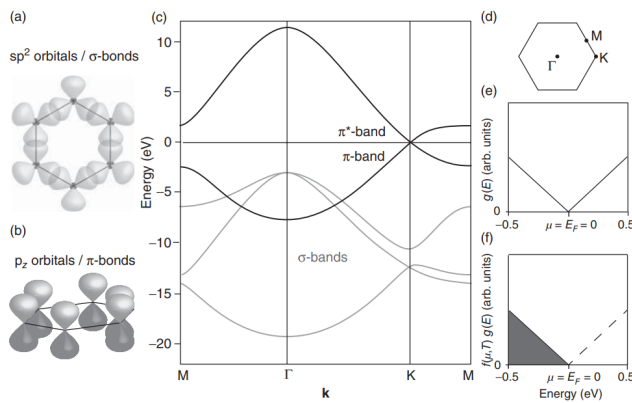


Figure 6.15 Origin of the electronic energy bands for graphene. (a) sp^2 hybrid orbitals giving rise to σ bonds. (b) p_z orbitals giving rise to π bonds. (c) Band structure taken from Kogan and Nazarov (2012) (not all the unoccupied bands are shown). (d) Two-dimensional first Brillouin zone. (e) Density of states in the immediate vicinity of Fermi energy. (f) Occupied density of states at $T = 0$ K (gray area).

- 8) *Tight-binding model:* (a) Show that the dispersion from Eq. (6.66) for the s-band in the tight-binding model can be approximated by a parabolic dispersion in the vicinity of $k = 0$, as in the nearly-free electron model. (b) Calculate the effective mass in this case, and discuss how it is related to the transfer integral γ_s that describes the ease of electrons moving from site to site.

- a) The dispersion is given by

$$E_s(k) = E_s - \beta_s - 2\gamma_s \cos(ka)$$

We can use a Taylor expansion to approximate the cosine function

$$\cos(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \dots$$

When $x \ll 1$ then $\cos(x) \approx 1 - \frac{x^2}{2}$. Using this we obtain the dispersion

$$E_s(k) = E_s - \beta_s - 2\gamma_s \left(1 - \frac{k^2 a^2}{2}\right) = E_s - \beta_s - 2\gamma_s + \gamma_s k^2 a^2$$

This is exactly a parabolic dispersion as expected

b) The effective mass is defined as

$$m^* = \hbar^2 \left(\frac{d^2 E(k)}{dk^2} \right)^{-1}$$

We just need to take the double derivative so let us quickly do that

$$\frac{dE}{dk} = 2\gamma_s a^2 k$$

$$\frac{d^2 E}{dk^2} = 2\gamma_s a^2$$

This gives us as effective mass of

$$m^* = \frac{\hbar^2}{2\gamma_s a^2}$$

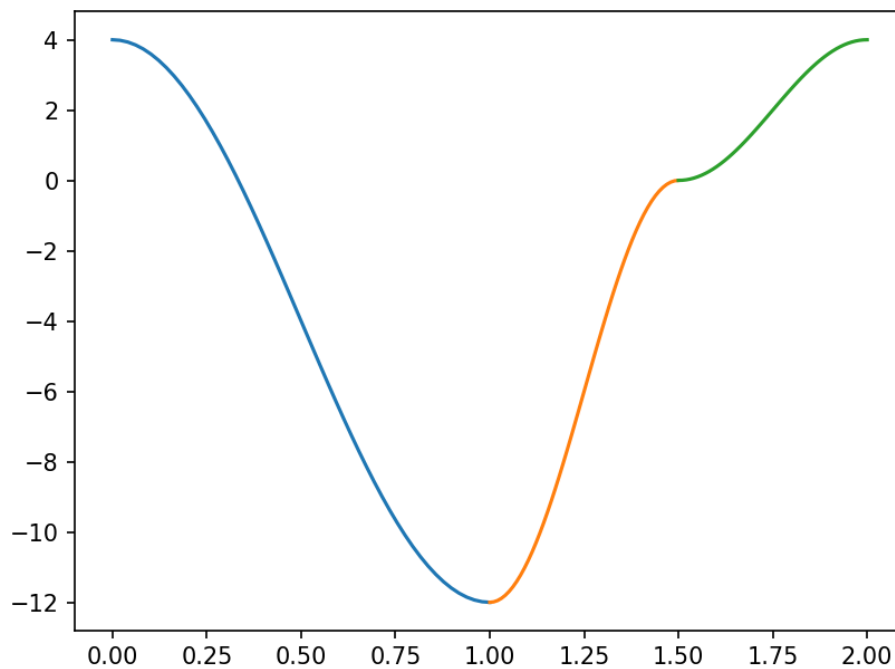
Notice that the effective mass and transfer integral is inverse proportional. This means that a small effective mass

10) *Three-dimensional band structures:* We have calculated the electronic dispersion for a band derived from s orbitals in one dimension. The result is given in Eq. (6.66). Due to the spherical symmetry of s or-

bitals, the band structure is still quite simple for three-dimensional lattices. For an fcc lattice, the energy of an isolated s band as a function of the three-dimensional wave vector $\mathbf{k} = (k_x, k_y, k_z)$ is given by

$$E(\mathbf{k}) = -4\gamma_s \left[\cos \frac{k_x a}{2} \cos \frac{k_y a}{2} + \cos \frac{k_z a}{2} \cos \frac{k_y a}{2} + \cos \frac{k_z a}{2} \cos \frac{k_x a}{2} \right], \quad (6.73)$$

where a is the fcc lattice constant and we have ignored the energy offset $E_s - \beta$. For a derivation of this, see for example Ashcroft and Mermin, Section 6.9. Plot this band structure in the same way as that of aluminum in Figure 6.12b, but only along X- Γ -L. Hint: The coordinates of these high symmetry points are: X = $(2\pi/a)(1, 0, 0)$; $\Gamma = (0, 0, 0)$; L = $(2\pi/a)(1/2, 1/2, 1/2)$.



```
import numpy as np
import matplotlib.pyplot as plt

def function(x,y,z):
```

```
Part1 = (np.cos(x/2))*(np.cos(y/2))
Part2 = (np.cos(z/2))*(np.cos(y/2))
Part3 = (np.cos(x/2))*(np.cos(z/2))
return -4*(Part1 + Part2 + Part3)

X_Gamma = np.linspace(1, 0, 50)
Gamma_L = np.linspace(0, 0.5, 50)
L_X_kx = np.linspace(0.5, 1, 50)
L_X_ky_kz = np.linspace(0.5, 0, 50)

Part1 = np.linspace(0, 1, 50)
Part2 = np.linspace(1, 1.5, 50)
Part3 = np.linspace(1.5, 2, 50)

fig, ax = plt.subplots()
ax.plot(Part1, function(2*np.pi*X_Gamma, 0, 0))
ax.plot(Part2, function(2*np.pi*Gamma_L, 2*np.pi*Gamma_L, 2*np.pi*Gamma_L))
ax.plot(Part3, function(2*np.pi*L_X_kx, 2*np.pi*L_X_ky_kz, 2*np.pi*L_X_ky_kz))

plt.show()
```

Problems 13

- 11) *Transport properties:* Show, in one dimension, that the average group velocity for a filled band is zero. Hint: For a chain with lattice constant a and macroscopic length L , the distance between the allowed k values is $2\pi/L$ – that is, it is very small. It is then useful to write the sum over the allowed k -points as an integral.

The group velocity is given by

$$v_g = \frac{d\omega(k)}{dk} = \frac{1}{\hbar} \frac{dE(k)}{dk}$$

The allowed k -points are

$$k = \frac{2\pi n}{aN} = \frac{2\pi}{L} n$$

We only need to find the average within the Brillouin zone because of periodicity. The first Brillouin zone is from $k = -\frac{\pi}{a}$ to $k = \frac{\pi}{a}$. The number of allowed k -points within the Brillouin zone is N if there are N cells. The average group velocity is given by

$$\langle v_g \rangle = \frac{1}{N} \sum_{k=-\frac{\pi}{a}}^{\frac{\pi}{a}} v_g(k) = \frac{a}{2\pi} \sum_{k=-\frac{\pi}{a}}^{\frac{\pi}{a}} v_g(k) \Delta k$$

Because the distance between allowed k values are so small we can turn this sum into an integral and we can calculate

$$\begin{aligned} \langle v_g \rangle &= \frac{a}{2\pi} \int_{-\frac{\pi}{a}}^{\frac{\pi}{a}} v_g(k) dk = \frac{a}{2\pi\hbar} \int_{-\frac{\pi}{a}}^{\frac{\pi}{a}} \frac{dE(k)}{dk} dk = \frac{a}{2\pi\hbar} [E(k)]_{-\frac{\pi}{a}}^{\frac{\pi}{a}} \\ &= \frac{a}{2\pi\hbar} \left[E\left(\frac{\pi}{a}\right) - E\left(-\frac{\pi}{a}\right) \right] \end{aligned}$$

Notice that the dispersion relation is periodic

$$\omega(k) = \omega\left(k + \frac{2\pi}{a}\right)$$

This means that

$$E(k) = E\left(k + \frac{2\pi}{a}\right)$$

Therefore

$$\langle v_g \rangle = \frac{1}{\hbar N} \left[E\left(\frac{\pi}{a}\right) - E\left(-\frac{\pi}{a}\right) \right] = 0$$

- 12) *Transport properties:* Show that the effective mass for free electrons is equal to the mass m_e of a free electron.

The effective mass is defined as

$$m^* = \hbar^2 \left(\frac{d^2 E(k)}{dk^2} \right)^{-1}$$

We know the energy for the free electron is

$$E = \frac{\hbar^2 |\vec{k}|^2}{2m_e}$$

$\Downarrow$

$$\frac{d^2 E(k)}{dk^2} = \frac{d^2}{dk^2} \frac{\hbar^2 k^2}{2m_e} = \frac{\hbar^2}{m_e} \frac{dk}{dk} = \frac{\hbar^2}{m_e}$$

Inserting this into the formula gives

$$m^* = \hbar^2 \left(\frac{\hbar^2}{m_e} \right)^{-1} = m_e$$

- 5) *Doped semiconductors:* (a) Calculate the effective number of states per volume, N_{eff}^c , for the conduction band of silicon at $T = 150$ K and $T = 300$ K. (b) In the intermediate-temperature (extrinsic) case of Figure 7.7a, it appears that almost all the electrons in the donor atoms have been excited into the conduction band. On the other hand, the chemical potential in this situation must be close to the middle of the gap, which means that the Fermi–Dirac distribution is small at the donor levels but even smaller in the conduction band. So, why are the

electrons in the conduction band and not in the donor levels? (Hint: Keep in mind that the doping concentration is on the order 10^{20} m^{-3}).

a) Calculations

Table 7.3 Effective masses for some semiconductors.

Material	m_e^*/m_e	m_h^*/m_e
Ge	0.60	0.28
Si	0.43	0.54
CdSe	0.13	0.45
InSb	0.015	0.39
InAs	0.026	0.41
GaAs	0.065	0.50

$$N_{\text{eff}}^c = \frac{1}{\sqrt{2}} \left(\frac{m_e^* kT}{\pi \hbar^2} \right)^{\frac{3}{2}}$$

$$N_{eff}^C(150K) = \frac{1}{\sqrt{2}} \left(\frac{0.43 \cdot 9.109 \cdot 10^{-31} kg \cdot 1.381 \cdot 10^{-23} \frac{J}{K} \cdot 150K}{\pi \cdot (1.055 \cdot 10^{-34} J \cdot s)^2} \right)^{\frac{3}{2}} = 0,002499418 \left(\frac{1}{nm^3} \right)$$

$$= 2,49942 \cdot 10^{24} m^{-3}$$

$$N_{eff}^C(300K) = \frac{1}{\sqrt{2}} \left(\frac{0.43 \cdot 9.109 \cdot 10^{-31} kg \cdot 1.381 \cdot 10^{-23} \frac{J}{K} \cdot 300K}{\pi \cdot (1.055 \cdot 10^{-34} J \cdot s)^2} \right)^{\frac{3}{2}}$$

$$= 0,0070694227092764 \left(\frac{1}{nm^3} \right) \approx 7,06942 \cdot 10^{24} m^{-3}$$

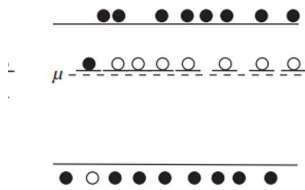
- b) You have to consider the Fermi-Dirac distribution and the population donor atoms and the amount of states in the conduction and.

At higher temperatures then the Fermi-Dirac distribution becomes wider. Furthermore the number of atoms also depend on the density of states.

$$N = \int_0^{\infty} g(\epsilon) \bar{n}_{FD}(\epsilon) d\epsilon$$

The number of atoms in the conduction band and donor level has to match.

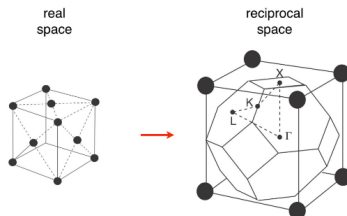
Intermediate temperature (extrinsic)



In this case $\bar{n}_{FD}(\epsilon)$ is small in the conduction band but $g(\epsilon)$ is very large compared to the donor level. This evens out for the donor level where $\bar{n}_{FD}(\epsilon)$ is large instead while $g(\epsilon)$ is very small. That is why the donor level is so empty despite the chemical potential being very close to the donor level

- 8) *Si band structure*: Silicon is an example of a semiconductor with an indirect band gap. The valence band maximum is found at the Γ point of the Brillouin zone and the conduction band minimum is placed along the Γ -X symmetry line, as seen in Figure 6.13. Due to the symmetry of the Brillouin zone (see sketch in Figure 6.12), there are multiple degenerate conduction band minima. How many? For the hypothetical case of the conduction band minimum placed *exactly* at the X point, how many degenerate minima would there be?

In the case of Si then the conduction band minimum is somewhere between $\Gamma - \chi$ which means that there are 6 degenerate conduction band due to centrosymmetry. In other words you can draw 6 $\Gamma - \chi$ lines in the reciprocal space



If the minimum was exactly at point X then the minimum would be 3-fold degenerate. You can technically draw 6 X points in the figure above but remember that those points are shared with the next cell, which means that you in reality only have 3 unique X-points.

Think of it like material chemistry where you have to count how many atoms are in your unit cell when you have an atom placed on each surface of a cube. Even though you can draw 6, you only have 3 atoms in your unit cell in reality. The same concept applies here

- 7) **Doped semiconductors:* Figure 7.7 shows the temperature dependences of the chemical potential and electron density for an n-doped semiconductor. Numerically generate a corresponding plot based on the following data: The semiconductor shall be Si with a band gap of 1.1 eV and effective masses of $0.43 m_e$ and $0.54 m_e$ for the electrons and holes, respectively. The donor concentration shall be 10^{22} m^{-3} . There is also an additional acceptor density of 10^{19} m^{-3} due to contaminations in the crystal. (Such contaminations can, for example, arise accidentally during the crystal growth process.) The acceptor and donor levels are 50 meV above (below) the valence band maximum (conduction band minimum). There are a number of details influencing this calculation that we have not discussed in our simple treatment. Therefore, you should merely expect a qualitative agreement with Figure 7.7. However, the general behavior of the chemical potential and the different temperature regions (intrinsic, extrinsic and freeze-out) should be reproduced.

The law of mass action that applies in all cases

$$n_0 p_0 = 4 \left(\frac{kT}{2\pi\hbar^2} \right)^3 (m_e^* m_h^*)^{\frac{3}{2}} e^{-\frac{E_g}{kT}} = n_i^2$$

⇕

$$n_0 = \frac{n_i^2}{p_0}$$

Charge neutrality requires that

$$n_0 + N_a^- = p_0 + N_d^+$$

Where N_a^- is the number of acceptors that have taken in an electron and N_d^+ is the number of ionized donors. We can approximately describe the number of acceptors with the Fermi-Dirac distribution even though the book says it is a bad idea. This marks our first approximation. Remember the two spin states when you derive the grand partition function for each energy level

$$Z = 1 + 2e^{-\frac{(\epsilon - \mu)}{kT}}$$

↓

$$n_{FD} = \frac{1}{1 + \frac{1}{2}e^{-\frac{(\epsilon - \mu)}{kT}}}$$

The total number of ionized acceptors must be given by the Fermi-Dirac distribution multiplied by the total acceptor density

$$N_a^- = \frac{N_a}{1 + \frac{1}{2}e^{-\frac{(E_a - \mu)}{kT}}}$$

Similarly the number of ionized donors is given by the following

$$1 - n_{FD} = 1 - \frac{1}{1 + \frac{1}{2}e^{-\frac{(\epsilon - \mu)}{kT}}} = \frac{1 + \frac{1}{2}e^{-\frac{(\epsilon - \mu)}{kT}} - 1}{1 + \frac{1}{2}e^{-\frac{(\epsilon - \mu)}{kT}}} = \frac{1}{2e^{-\frac{(\epsilon - \mu)}{kT}} + 1}$$

Same logic applies for the total number of ionized donors

$$N_d^+ = \frac{N_d}{2e^{-\frac{(E_d - \mu)}{kT}} + 1}$$

Great. We now have two formulas and two unknowns that we can solve

$$n_0 = \frac{n_i^2}{p_0}$$

$$n_0 + N_a^- = p_0 + N_d^+$$

↓

$$\frac{n_i^2}{p_0} + N_a^- = p_0 + N_d^+$$

⇕

$$n_i^2 + N_a^- p_0 = p_0^2 + N_d^+ p_0$$

⇕

$$p_0^2 + p_0(N_d^+ - N_a^-) - n_i^2 = 0$$

The solution to this equation is the quadratic solution

$$p_0 = \frac{-(N_d^+ - N_a^-) \pm \sqrt{(N_d^+ - N_a^-)^2 + 4n_i^2}}{2}$$

Let us consolidate. We know that we can find the electron density from the law of mass action as

$$n_0 = \frac{n_i^2}{p_0}, \quad n_i^2 = 4 \left(\frac{kT}{2\pi\hbar^2} \right)^3 (m_e^* m_h^*)^{\frac{3}{2}} e^{-\frac{E_g}{kT}}$$

The hole density we just derived from the law of charge neutrality

$$p_0 = \frac{-(N_d^+ - N_a^-) \pm \sqrt{(N_d^+ - N_a^-)^2 + 4n_i^2}}{2}$$

Only the positive solution makes sense because a negative hole density is unphysical

$$p_0 = \frac{-(N_d^+ - N_a^-) + \sqrt{(N_d^+ - N_a^-)^2 + 4n_i^2}}{2}$$

The ionized donor and acceptor density we found from the Fermi-Dirac distribution

$$N_a^- = \frac{N_a}{1 + \frac{1}{2} e^{\frac{(E_a - \mu)}{kT}}}, \quad N_d^+ = \frac{N_d}{2e^{-\frac{(E_d - \mu)}{kT}} + 1}$$

We now have one issue and that is finding a formula for the chemical potential. This is where we have to start thinking numerically. We know from the statistical physics that the chemical potential is not allowed to create atoms out of the blue so this equation has to hold

$$N = \int_{-\infty}^{\infty} g(\epsilon) n_{FD}(\epsilon) d\epsilon$$

Defining the valence band maximum as the point of zero energy and working in “per unit volume” gives

$$\frac{N}{V} = \int_{-\infty}^0 g_{VB}(\epsilon) n_{FD}(\epsilon) d\epsilon + N_a n_{FD}(E_a) + N_d n_{FD}(E_d) + \int_{E_g}^{\infty} g_{CB}(\epsilon) n_{FD}(\epsilon) d\epsilon$$

For parabolic bands the density of states per unit volume can be described from the free electron model

$$g(E) = \frac{1}{2\pi^2} \left(\frac{2m}{\hbar^2} \right)^{\frac{3}{2}} \sqrt{E}$$

This means that the valence band can be described as

$$g_{VB}(\epsilon) = \frac{1}{2\pi^2} \left(\frac{2m_h^*}{\hbar^2} \right)^{\frac{3}{2}} \sqrt{E_V - E}$$

And the conduction band as

$$g_{CB}(\epsilon) = \frac{1}{2\pi^2} \left(\frac{2m_e^*}{\hbar^2} \right)^{\frac{3}{2}} \sqrt{E - E_C}$$

Now the final question we have to ask ourselves is what do we choose for N/V ? For this we can consider the lower bound of N . At $T=0$ then the conduction band should be empty and the valence band should be totally filled. Furthermore, the electrons from the donor level will also fill the acceptor level. However since the concentration of donor electrons is much larger than the acceptor level then both the donor level and acceptor level will effectively be totally filled.

So that means that $\frac{N}{V} > \frac{10^{22}}{m^3}$.

For the keen eye our problem is actually from

$$\int_{-\infty}^0 g_{VB}(\epsilon) n_{FD}(\epsilon) d\epsilon$$

We do not have a lower bound so in theory N/V should be infinity. We therefore have to choose our lower bound by placing the chemical potential fixed in the middle of our band gap and going to the highest temperature that we want to plot from. Then we have to see in what region the FD distribution changes. This marks the upper bound for the lower integration limit. We then choose our integration limit x from this knowledge and calculate for $T=0$ the following integral

$$\int_x^0 g_{VB}(\epsilon) d\epsilon$$

Finally, we choose

$$\frac{N}{V} = \int_x^0 g_{VB}(\epsilon) d\epsilon + \frac{10^{22}}{m^3}, \quad T = 0$$

With this we now have a way to numerically calculate the chemical potential.

Okay let us summarize what we have to program

1. We have to make a function that calculates n_i^2 as a function of temperature with the formula

$$n_i^2 = 4 \left(\frac{kT}{2\pi\hbar^2} \right)^3 (m_e^* m_h^*)^{\frac{3}{2}} e^{-\frac{E_g}{kT}}$$

2. We have to make two functions that calculates the DOS for the conduction band and valence band as a function of energy with the definition that the valence band maximum E_V is at zero energy

$$g_{VB}(\epsilon) = \frac{1}{2\pi^2} \left(\frac{2m_h^*}{\hbar^2} \right)^{\frac{3}{2}} \sqrt{E_V - E}, \quad E_V = 0$$

$$g_{CB}(\epsilon) = \frac{1}{2\pi^2} \left(\frac{2m_e^*}{\hbar^2} \right)^{\frac{3}{2}} \sqrt{E - E_C}, \quad E_C = E_g$$

3. We have to calculate the Fermi-Dirac distribution as a function of energy and temperature when the chemical potential is in the middle of the band gap

$$n_{FD}(\epsilon, \mu, T) = \frac{1}{1 + \frac{1}{2} e^{\frac{(\epsilon - \mu)}{kT}}}, \quad \mu = \frac{E_g}{2}$$

4. We have to choose a lower integration limit, x, from our evaluation in step 3 and calculate N/V

$$\frac{N}{V} = \int_x^0 g_{VB}(\epsilon) d\epsilon + \frac{10^{22}}{m^3}$$

5. We have to set up an equation that can calculate the chemical potential for a series of temperatures using the following expression

$$\frac{N}{V} = \int_x^0 g_{VB}(\epsilon) n_{FD}(\epsilon, \mu, T) d\epsilon + N_a n_{FD}(E_a, \mu, T) + N_d n_{FD}(E_d, \mu, T) + \int_{E_g}^{\infty} g_{CB}(\epsilon) n_{FD}(\epsilon, \mu, T) d\epsilon$$

6. We define the ionized donor and acceptor density as a function of chemical potential and temperature

$$N_a^-(\mu, T) = \frac{N_a}{1 + \frac{1}{2} e^{\frac{(E_a - \mu)}{kT}}}, \quad N_d^+(\mu, T) = \frac{N_d}{2e^{-\frac{(E_d - \mu)}{kT}} + 1}$$

7. We define the hole density using the previous functions as

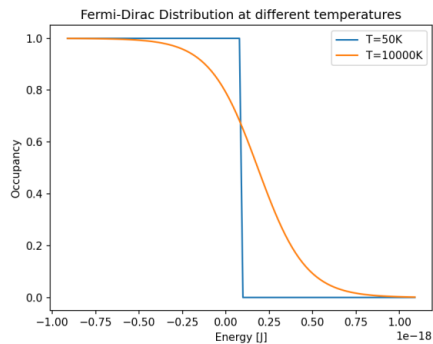
$$p_0(\mu, T) = \frac{-(N_d^+ - N_a^-) + \sqrt{(N_d^+ - N_a^-)^2 + 4n_i^2}}{2}$$

8. We define the electron density using the previous functions as

$$n_0(\mu, T) = \frac{n_i^2(T)}{p_0(\mu, T)}$$

9. We can now plot the electron density as a function of temperature using all of the above information

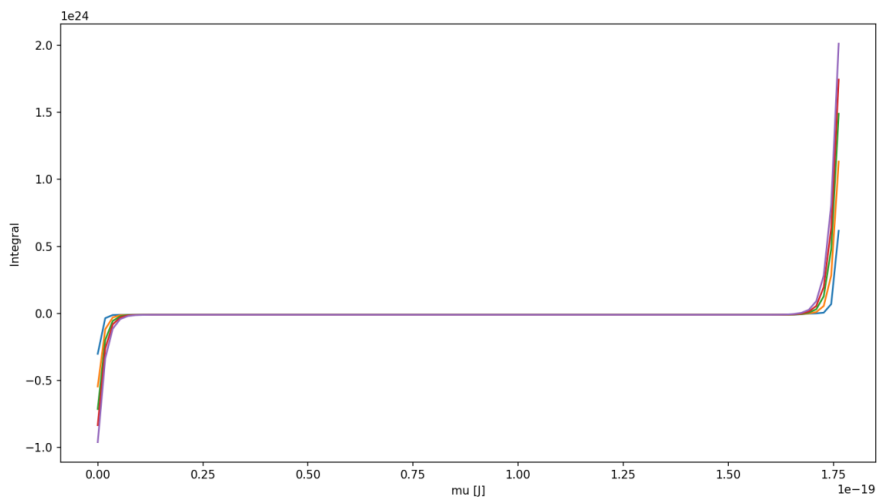
I've done some programming and plotted the Fermi-Dirac distribution and can say with confidence that our lower integration limit x should be $-10^{-17} J$



I plot the following equation and plot it as a function of μ at 50K, 100K and some temperatures in-between to try and see when the function goes to zero

$$0 = \int_{-10^{-17}}^0 g_{VB}(\epsilon) n_{FD}(\epsilon, \mu, T) d\epsilon + N_a n_{FD}(E_a, \mu, T) + N_d n_{FD}(E_d, \mu, T) + \int_{E_g}^{10^{-17}} g_{CB}(\epsilon) n_{FD}(\epsilon, \mu, T) d\epsilon - \frac{N}{V}$$

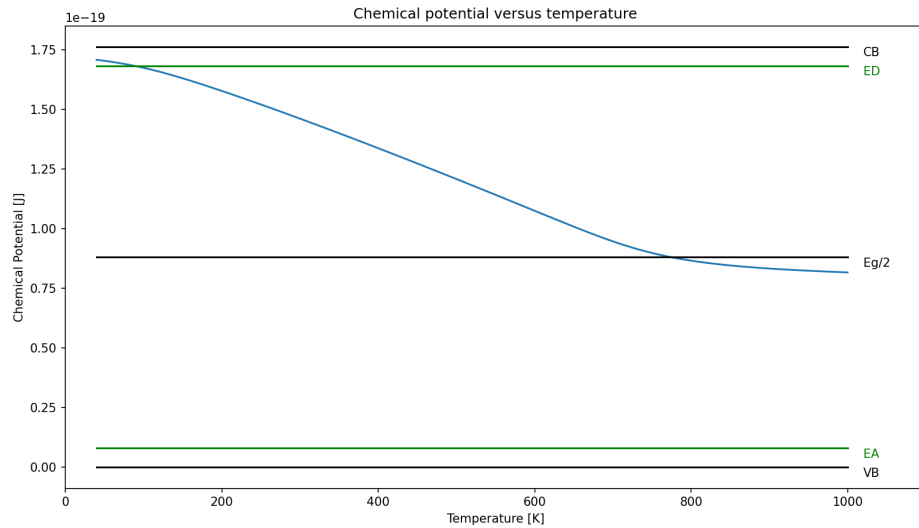
It may look like it is zero in a whole interval but zooming in shows that it is actually only the far right part which happens to be when μ is close to the conduction band. The result looks a bit jagged because the numerical calculations are quite heavy



My guess for μ that I should use to solve my integral with fsolve is around $1.73 \cdot 10^{-19}J$ for a temperature of 50K. I can now calculate the chemical potential as a function of temperature and this is what I get. The chemical potential is initially above the donor level as expected since the donor concentration is much larger than the acceptor concentration. However, interestingly the chemical potential goes slightly below half the band gap. I actually expect it to be slightly above half the band gap in the intrinsic state because $m_h > m_e$ in this formula

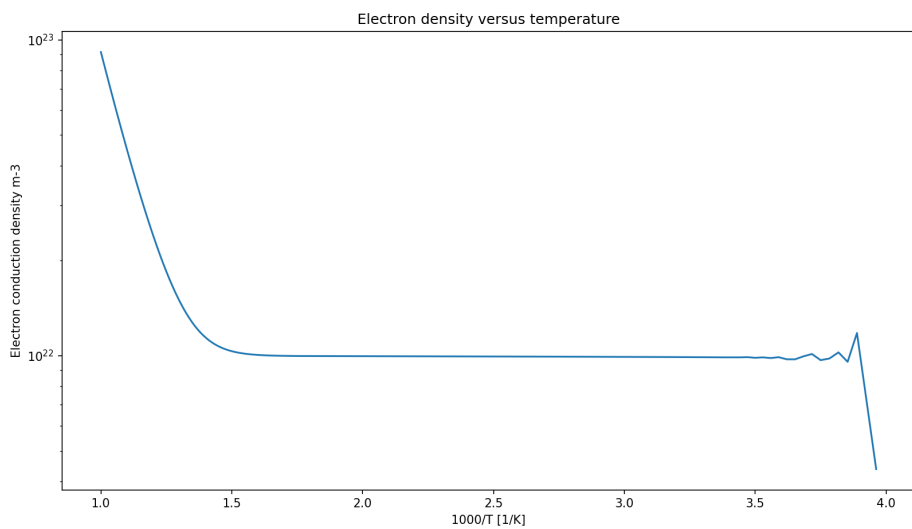
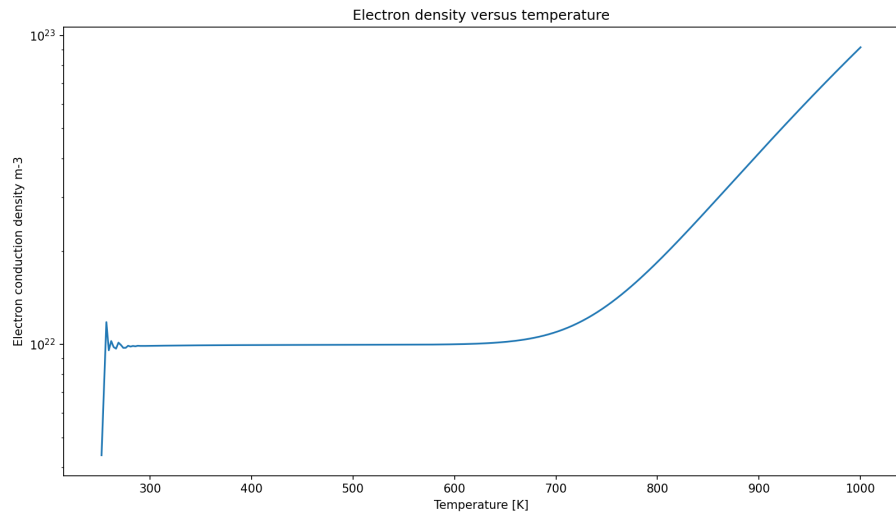
$$\mu = \frac{E_g}{2} + \frac{3}{4}kT \ln\left(\frac{m_h^*}{m_e^*}\right)$$

But the deviation is not so large. Chances are that it is due to our numerical approximations that it behaves like this. I do get errors and overflow warnings in my code so that could be the source of the error. But still we get something close to what were supposed to get



And the resulting electron density as a function of temperature is as expected and we clearly see our three temperature regions. It can be a bit difficult to calculate the low temperature end because of numerical instability as you can see pretty clearly. But qualitatively it behaves as expected with a freeze-out, extrinsic and intrinsic region. And that concludes the problem.

I have to be honest this took me 6 hours to solve and started from scratch 3 times to try different approaches and this is the best I could get. I am sure there are better approximations or solutions that can be found but in our case, I would say this is “good enough”



```
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import quad
from scipy.optimize import fsolve

m_electron = 9.1093837*10**(-31) #kg
C = 1.60217663*10**(-19) #C
k= 1.381*10**(-23)
hbar = 1.055*10**(-34)
```

```

gap = 1.1*C
me = 0.43*m_electron
mh = 0.54*m_electron

# Energy of donor level is the band gap minus the 50 meV
ED = gap-50*10**(-3)*C
ND = 10**22

EA = 50*10**(-3)*C
NA = 10**19

# Lower integration limit evaluated by the Fermi-Dirac distribution
xint = -10**-17

# Formula for ni^2
def ni2(T):
    express = 4*(k*T/(2*np.pi*hbar**2))**3 * (me*mh)**(3/2)*np.exp(-gap/(k*T))
    return express

# Formula for DOS in VB and CB
def gvb(E):
    return 1/(2*np.pi**2) * (2*mh/(hbar**2))**(3/2)*np.sqrt(0-E)

def gcb(E):
    return 1/(2*np.pi**2) * (2*me/(hbar**2))**(3/2)*np.sqrt(E-gap)

# Formula for ionized donor and acceptor density
def INa(mu, T):
    return NA/(1+1/2*np.exp((EA-mu)/(k*T)))

def INd(mu, T):
    return ND/(1+2*np.exp(-(ED-mu)/(k*T)))

# Formula for hole density
def p0(mu, T):
    return (-(INd(mu, T)-INa(mu, T))+np.sqrt((INd(mu, T)-INa(mu, T))**2 +
4*ni2(T)))/2

# Formula for electron density
def n0(mu, T):
    return ni2(T)/p0(mu, T)

# Formula for FD distribution
def nFD(E, mu, T):
    return 1/(1+1/2*np.exp((E-mu)/(k*T)))

def NV(x):
    integral = lambda x: gvb(x)

```

```

        return quad(integral, x, 0)[0] + ND

def big_equation(mu, T):
    integral1 = lambda x: gvb(x)*nFD(x, mu, T)
    integral2 = lambda x: gcb(x)*nFD(x, mu, T)
    return quad(integral1, xint, 0)[0] + NA*nFD(EA, mu, T) + ND*nFD(ED, mu, T)
+ quad(integral2, gap, -xint)[0] - NV(xint)

fig, ax = plt.subplots()
Es = np.linspace(-10**-18+gap/2, 10**-18+gap/2, 100)
ax.plot(Es, nFD(Es, gap/2, 50))
ax.plot(Es, nFD(Es, gap/2, 10000))

ax.set_xlabel('Energy [J]')
ax.set_ylabel('Occupancy')
ax.set_title('Fermi-Dirac Distribution at different temperatures')
ax.legend(['T=50K', 'T=10000K'])
# plt.show()

# Use this plot to find a good initial guess to the solution of the integral.
It should cross zero
fig, ax = plt.subplots()
mu_xs = np.linspace(0, gap, 100)
ax.plot(mu_xs, [big_equation(i, 50) for i in mu_xs])
ax.plot(mu_xs, [big_equation(i, 75) for i in mu_xs])
ax.plot(mu_xs, [big_equation(i, 90) for i in mu_xs])
ax.plot(mu_xs, [big_equation(i, 100) for i in mu_xs])
ax.plot(mu_xs, [big_equation(i, 110) for i in mu_xs])

ax.set_xlabel('mu [J]')
ax.set_ylabel('Integral')
# plt.show()

Ts = np.linspace(50, 1000, 400)
mu_cal = []
guess = 1.73*10**-19
for i in Ts:
    chemical_potential = fsolve(lambda x: big_equation(mu=x, T=i), x0=guess,
maxfev=1000, xtol=1e-22)
    guess = chemical_potential
    mu_cal.append(chemical_potential)

fig, ax = plt.subplots()
ax.plot(Ts, mu_cal)
ax.plot([Ts[0], Ts[-1]], [0, 0], color='black')
ax.plot([Ts[0], Ts[-1]], [gap/2, gap/2], color='black')
ax.plot([Ts[0], Ts[-1]], [gap, gap], color='black')
ax.text(Ts[-1]+20, 0-EA/2, 'VB')

```

```

ax.text(Ts[-1]+20, gap/2-EA/2, 'Eg/2')
ax.text(Ts[-1]+20, gap-EA/2, 'CB')

ax.plot([Ts[0], Ts[-1]], [ED, ED], color='g')
ax.plot([Ts[0], Ts[-1]], [EA, EA], color='g')
ax.text(Ts[-1]+20, ED-EA/2, 'ED', color='g')
ax.text(Ts[-1]+20, EA/2, 'EA', color='g')

ax.set_xlabel('Temperature [K]')
ax.set_ylabel('Chemical Potential [J]')
ax.set_title('Chemical potential versus temperature')
ax.set_xlim([0, 1100])
plt.show()

fig,ax = plt.subplots()

ys = [n0(i, j) for i,j in zip(mu_cal, Ts)]
ax.semilogy(Ts, ys)

ax.set_ylabel('Electron conduction density m-3')
ax.set_xlabel('Temperature [K]')
ax.set_title('Electron density versus temperature')
# plt.show()

fig,ax = plt.subplots()

ys = [n0(i, j) for i,j in zip(mu_cal, Ts)]
ax.semilogy(1000/Ts, ys)

ax.set_ylabel('Electron conduction density m-3')
ax.set_xlabel('1000/T [1/K]')
ax.set_title('Electron density versus temperature')
plt.show()

```

Problems 14

- 3) *Ferromagnetic ordering*: For the ferromagnetic elements Gd^{3+} (with seven 4f electrons) and Fe^{2+} (with six 3d electrons), calculate the highest possible magnetization per atom and compare it to the experimental values measured at ≈ 0 K, which are $7.6\mu_B$ and $2.2\mu_B$, respectively. Discuss the results.

Magnetization is given by

$$\vec{M} = \frac{N}{V} \vec{\mu}$$

The magnetization per atom is therefore just the magnetic moment which in the case with spin-orbit coupling is described by

$$\mu_J = -g_J m_J \mu_B, \quad -J, -J+1, \dots, J-1, J$$

The highest possible magnetization is the one with maximum μ which is when $m_J = -J$ such that

$$\mu_{-J} = g_{-J} J \mu_B$$

The Lande splitting factor has to be calculated with

$$L = \sum m_l, \quad S = \sum m_s$$

Gd^{3+} has 7 unpaired electrons which means that the total spin is

$$S = 7 \cdot \frac{1}{2} = \frac{7}{2}$$

Therefore $L = 0$ and $J = S$

$$\mu_J = -g_J m_J \mu_B, \quad m_J = \frac{7}{2}, \frac{5}{2}, \dots, -\frac{5}{2}, -\frac{7}{2}$$

And

$$g_J = \frac{3J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}$$

$$g_J = \frac{3\left(\frac{7}{2}\right)\left(\left(\frac{7}{2}\right)+1\right) + \frac{7}{2}\left(\frac{7}{2}+1\right)}{2\left(\frac{7}{2}\right)\left(\left(\frac{7}{2}\right)+1\right)} = 2$$

The highest possible magnetization is the one with most negative m_J

$$\mu_{-\frac{7}{2}} = -2\left(-\frac{7}{2}\right)\mu_B = 7\mu_B$$

Fe²⁺ has 4 unpaired electron which means

$$S = \frac{4}{2} = 2$$

The orbital angular momentum varies with $m_l = 0, \pm 1, \pm 2, \dots, \pm l$ where l is the orbital in question. For iron the electrons are distributed in the d-orbital which means that $l = 2$. There are 6 electrons in the d-orbital because the 2 electrons removed from Fe⁰ came from the 4s orbital. The electrons are placed in the d-orbitals to maximize L

$$L = 2 \cdot 2 + 1 + 0 - 1 - 2 = 2$$

And

$$J = L + S = 4$$

And

$$g_J = \frac{3J(J+1) + S(S+1) - L(L+1)}{2J(J+1)} = \frac{3 \cdot 4 \cdot (4+1) + 2(2+1) - 2(2+1)}{2 \cdot 4 \cdot (4+1)} = \frac{3}{2}$$

Therefore

$$\mu_J = -\frac{3}{2}(-4)\mu_B = 6\mu_B$$

Localized magnetic moments are clearly not a good description 3d transition metals

6) *Ferromagnetic ordering*: Starting from the Heisenberg model for localized spins, we derived the temperature-dependent magnetization of a two-state system in Eq. (8.42). The trouble with the expression is

that it has no simple analytical solution apart from the trivial solution $M(T) = 0$. Using a numerical approach, show that a nontrivial solution exists below the Curie temperature Θ_C , and that the temperature-dependent magnetization corresponds to that shown in Figure 8.5.

The expression we want to find solutions for are the following

$$\frac{M(T)}{M(0)} = \tanh\left(\frac{M(T)}{M(0)} \frac{\Theta_C}{T}\right)$$

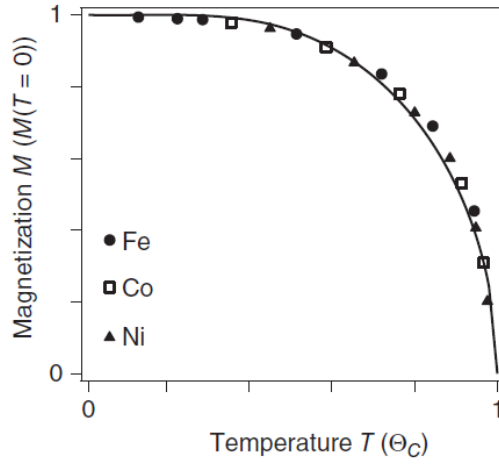


Figure 8.5 Temperature-dependent magnetization of Fe, Co, and Ni below the Curie temperature Θ_C . The line is the prediction according to (8.42) for a spin 1/2 system. The data points are taken from Tyler (1931).

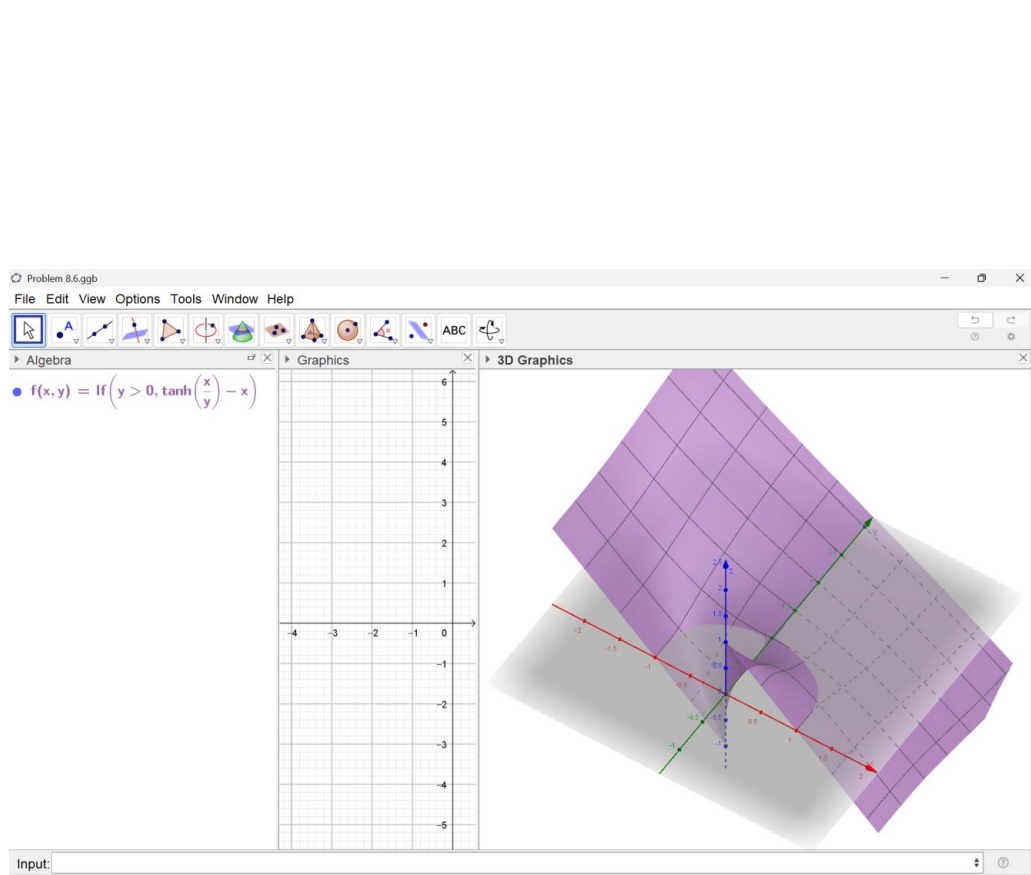
Let us create a function f such that the solutions to the equations are given when $f = 0$

$$f = \tanh\left(\frac{M(T) \Theta_C}{M(0) T}\right) - \frac{M(T)}{M(0)}$$

Since Θ_C and $M(0)$ are constant then let us just define them as 1. Since $M(T)$ is annoying to plot as a function of temperature let us just let f be a function of both $M(T)$ and T

$$f(M(T), T) = \tanh\left(\frac{M(T)}{T}\right) - M(T)$$

Here it is clear to see all the trivial solution along the y-axis with $x = M(T) = 0$ and then we have a ring of non-trivial solutions



7) *Curie–Weiss law*: Derive the Curie–Weiss law [Eq. (8.44)] and show that the Curie constant C corresponds to that in the Curie law [Eq. (8.29)].

The magnetization of a two state paramagnet was derived in early statistical physics

$$M(T) = \frac{\mu_B N}{V} \frac{e^x - e^{-x}}{e^x + e^{-x}}, \quad x = \frac{g_e |m_s| \mu_B (B_W + B_0)}{kT}$$

We know that the Weiss Field is defined as

$$B_W = \frac{M n_{nn} X V}{g_e^2 \mu_B^2 N}$$

And the magnetization when $T = 0$ is

$$M(0) = \frac{\mu_B N}{V}$$

This gives

$$\frac{M(T)}{M(0)} = \tanh(x)$$

Thus

$$\begin{aligned}
\frac{M(T)}{M(0)} &= \tanh\left(\frac{g_e |m_s| \mu_B (B_W + B_0)}{kT}\right) \\
&= \tanh\left(\frac{g_e |m_s| \mu_B B_W}{kT} + \frac{g_e |m_s| \mu_B B_0}{kT}\right) \\
&= \tanh\left(\frac{g_e |m_s| \mu_B}{kT} \frac{M n_{nn} X}{g_e^2 \mu_B^2 N} + \frac{g_e |m_s| \mu_B B_0}{kT}\right) \\
&= \tanh\left(\frac{|m_s|}{kT} \frac{M n_{nn} X}{g_e M(0)} + \frac{g_e |m_s| \mu_B B_0}{kT}\right)
\end{aligned}$$

For a two state paramagnet then

$$|m_s| = \frac{1}{2} = \frac{1}{g_e}$$

Therefore

$$\Theta_c = \frac{n_{nn} |m_s| X}{g_e k_B} = \frac{n_{nn} X}{g_e^2 k_B}$$

Which means

$$\frac{M(T)}{M(0)} = \tanh\left(\frac{M(T)}{M(0)} \frac{\Theta_c}{T} + \frac{g_e |m_s| \mu_B B_0}{kT}\right)$$

In the high temperature limit then we can use the approximation where $\tanh(x) \approx x$ for $|x| \ll 1$

$$\frac{M(T)}{M(0)} \approx \frac{M(T)}{M(0)} \frac{\Theta_c}{T} + \frac{g_e |m_s| \mu_B B_0}{kT}$$

The susceptibility is defined as

$$\chi_m = \frac{\mu_0 M}{B_0} \Leftrightarrow M = \frac{\chi_m B_0}{\mu_0}$$

Inserting this for $M(T)$ and inserting $M(0)$ gives

$$\frac{\chi_m B_0}{\mu_0 \frac{\mu_B N}{V}} = \frac{\chi_m B_0}{\mu_0 \frac{\mu_B N}{V}} \frac{\Theta_c}{T} + \frac{g_e |m_s| \mu_B B_0}{kT}$$

$\Updownarrow$

$$\frac{\chi_m B_0}{\mu_0 \frac{\mu_B N}{V}} \left(1 - \frac{\Theta_c}{T}\right) = \frac{g_e |m_s| \mu_B B_0}{kT}$$

$\Updownarrow$

$$\chi_m \left(\frac{T - \Theta_c}{T} \right) = \frac{N g_e |m_s| \mu_B^2 \mu_0}{V k T}$$

$\Downarrow$

$$\chi_m = \frac{\frac{N g_e |m_s| \mu_B^2 \mu_0}{V k}}{T - \Theta_c} = \frac{C}{T - \Theta_c}$$

Where

$$C = \frac{N g_e |m_s| \mu_B^2 \mu_0}{V k} \approx \frac{N \mu_B^2 \mu_0}{V k}$$

Curie-Weiss law

$$\chi_m = \frac{C}{T - \Theta_c}, \quad C = \frac{\mu_0 N g_J^2 \mu_B^2 J(J+1)}{3 V k}$$

So our C is the same if $J = \frac{1}{2}$ and $g_J = 2$

2) *Dielectric function*: Estimate the static dielectric constant of NaCl. Use that NaCl has a density $\rho = 2170 \text{ kg m}^{-3}$, a lattice constant $a = 5.6 \text{ \AA}$, and a Young's modulus $Y = 40 \text{ GPa}$. The optical ϵ can be taken from Table 9.2, and the result can be compared to the static ϵ from the same table.

The formula we have to use here is

$$\epsilon_r(\omega) = \frac{N e^2}{V \epsilon_0 M} \frac{\omega_0^2 - \omega^2}{(\omega_0^2 - \omega^2)^2 + \eta^2 \omega^2} + \epsilon_{opt}$$

Since we are interested in the static dielectric constant then $\omega = 0$ and

$$\epsilon_s = \frac{N e^2}{V \epsilon_0 M \omega_0^2} + \epsilon_{opt}$$

We know that we can find the resonance frequency from the reduced mass and force constant

$$\omega_0 = \sqrt{\frac{2\gamma}{M}}$$

The force constant is given by Young-modulus

$$Y = \frac{\gamma}{a} \Leftrightarrow \gamma = Y a$$

Therefore

$$\omega_0^2 = \frac{2Ya}{M}$$

This gives

$$\epsilon_s = \frac{N}{V} \frac{e^2}{\epsilon_0 M \frac{2Ya}{M}} + \epsilon_{opt}$$

We can choose to look at one unit cell such that $V = a^3$ which gives

$$\epsilon_s = \frac{e^2}{\epsilon_0 2Ya^4} + \epsilon_{opt}$$

$$= \frac{(1.602 \cdot 10^{-19} C)^2}{2 \cdot 8.854 \cdot 10^{-12} \frac{F}{m} \cdot 40 \cdot 10^9 Pa \cdot (5.6 \cdot 10^{-10} m)^4} + 2.34 = 2.71$$

Material	Static ϵ	Optical ϵ
Diamond	5.68	5.66
NaCl	6.1	2.34

Thats far away!

Re: Problem 9.2



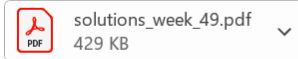
Simon Elliot Wall

To David Vu Bui



12/11/2025

You replied to this message on 13/11/2025 14.26.



Hi David,

To be honest, I did not look at the questions in a lot of detail. I took over the course this year from Philip and I just set the same questions. Philip also gets a much smaller number and I see that the difference is that Philip gets N/V from the density and masses rather than one oscillator per unit cell. I notice that if you look up NaCl on wikipedia it says there are 4 formula units per unit cell, this means that I think you should basically take the number you calculate and multiply by 4. $4/a^3$ gives the density given in the question.

In the note attached Philip argues why the number is small. The argument doesn't not really make sense to me. The claim is that the damping and lower resonance frequencies can make the value bigger at DC, but that is not the case and it is quite clear from the equation.

I calculated the resonant frequency of the bond using the equation you write and it comes out to be 3.4 THz, whereas the true value is around 5 THz. However, putting this in does not help because this will reduce the value not increase it.

Next time, I think I will not use this question! I think there are several issues with it. In section 4 of Philips book the relation for Youngs modulus and the spring constant is really in relation to a unit cell and not the inter-unit cell bonds (which is the case for an IR mode). It is related to the speed sound in the material so it is connected to the acoustic modes and not the optical ones.

However, if we ignore this fact. The length a is the distance between two Na (or Cl) ions and not the Na-Cl length, which is half this value. If I use $a/2$ instead of a and Philip's numbers of the density, I get 5.22

Therefore, giving the solution, I think it is best to use your method, but say that they should use $a/2$ rather than a , and not worry too much about the issue using Y for the spring constant.

I hope this helps

Simon